Chapter 21

Additional Conservative Esthetic Procedures

"Beauty is harmony of all parts, in whatsoever subject it appears, fitted together with such proportion and connection, that nothing could be added, diminished or altered…. But for the worse."

—Leon Battiste Alberte (1610)

Significant improvements in tooth-coloured restorative materials and adhesive techniques have resulted in numerous conservative esthetic treatment possibilities. Although restorative dentistry is a blend of art and science, conservative esthetic dentistry truly emphasizes the artistic component. As Goldstein stated, 'Esthetic dentistry is the art of dentistry in its purest form.'[1] As with many forms of art, conservative esthetic dentistry provides a means of artistic expression that feeds on creativity and imagination. Dentists find performing conservative esthetic procedures enjoyable, and patients appreciate the immediate esthetic improvements rendered, often without the need for local anaesthesia.

One of the greatest assets a person can have is a smile that shows beautiful, natural teeth (Fig. 21.1). Children and teenagers are especially sensitive about unattractive teeth. The restoration of a smile is one of the most appreciated and gratifying services a dentist can render. The positive psychological effects of improving a patient's smile often contribute to an *improved self-image* and *enhanced self-esteem*. These improvements make conservative esthetic dentistry particularly gratifying for the dentist and represent a new dimension of dental treatment for patients.

This chapter presents conservative esthetic procedures in the context of their clinical applications. The principles and clinical steps involved in adhesive bonding for the treatment alternatives discussed in this chapter are similar to those described in other chapters in this textbook. Only specific conservative esthetic clinical procedures or variations from previously described techniques are presented in this chapter.

Artistic Elements

Regardless of the result desired, certain basic artistic elements must be considered to ensure an optimal esthetic result. In conservative esthetic dentistry, these elements include the following:

- Shape or form
- Symmetry and proportionality
- Position and alignment
- Surface texture
- Colour
- Translucency

Shape or Form

The shape of teeth largely determines their esthetic appearance. To achieve optimal dental esthetics, it is imperative that natural anatomic forms be achieved. A basic knowledge of normal tooth anatomy is fundamental to the success of any conservative esthetic dental procedure.

Cosmetic Contouring

When viewing the clinical crown of an incisor from a facial (or lingual) position, the crown outline is trapezoidal. Subtle variations in shape and contour produce very different appearances, however. Rounded incisal angles, open incisal and facial embrasures and softened facial line angles typically characterize a youthful smile. A smile characteristic of an older individual having experienced attrition secondary to ageing typically exhibits incisal embrasures that are more closed and incisal angles that are more prominent (i.e. less rounded). Frequently, minor modification of existing tooth contours, sometimes referred to as *cosmetic contouring*, can effect a significant esthetic change (see section Alterations of Shape of Natural Teeth). Reshaping enamel by rounding incisal angles, opening incisal embrasures and reducing prominent facial line angles can produce a more youthful appearance (Fig. 21.2).

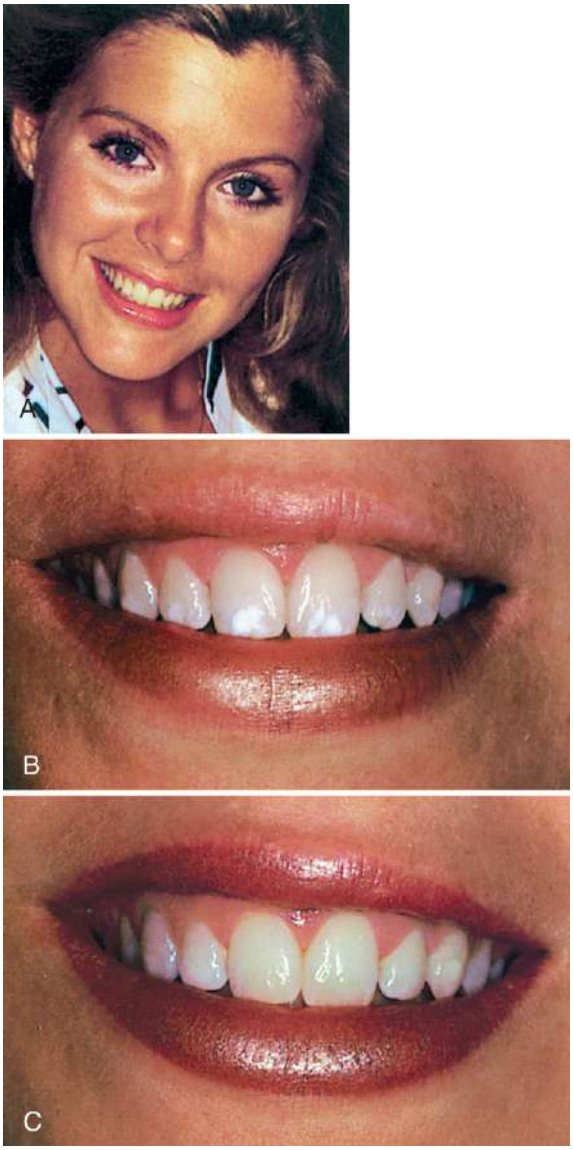


Figure 21.1 Examples of conservative esthetic procedures. (A) A beautiful radiant smile is one of the greatest assets a person can have; (B) the appearance of this aspiring young model was marred by hypocalcified areas of maxillary anterior teeth and (C) a simple treatment consisted of removing part of the discoloured enamel, acid etching the preparations and restoring with direct-composite partial veneers.
(Courtesy: Dr. C. L. Sockwell.)

Significant generalized esthetic changes are possible when treating all anterior teeth (and occasionally the first premolars) visible in the patient's smile. This fact is particularly true when placing full-coverage facial restorations such as veneers (see section Veneers). With this treatment method, the dentist can produce significant changes in tooth shapes and forms to yield a variety of different appearances.

Although less extensive, restoring an individual tooth rather than all anterior teeth simultaneously may require greater artistic ability. Generalized restoration of all anterior teeth with full facial veneers affords the dentist significant control of the contours generated. When treating an isolated tooth, however, the success of the result is determined largely by how well the restored tooth esthetically matches the surrounding natural teeth. The contralateral tooth to the one being restored should be examined closely for subtle characterizing features such as developmental depressions, embrasure form, prominences or other distinguishing characteristics of form. A high degree of realism must be reproduced artfully to achieve optimal esthetics when restoring isolated teeth or areas.

Illusions of Shape

- The border outline of an anterior tooth (i.e. facial view) is primarily two dimensional (i.e. length and width). The third dimension of depth is crucial, however, in creating illusions, especially those of apparent width and length.
- Prominent areas of contour on a tooth typically are highlighted with direct illumination, making them more noticeable, whereas areas of depression or diminishing contour are shadowed and less conspicuous. By controlling the areas of light reflection and shadowing, full facial coverage restorations (in particular) can be esthetically contoured to achieve various desired illusions of form.
- The apparent size of a tooth can be changed by altering the position of facial prominences or heights of contour without changing the actual dimension of the tooth.

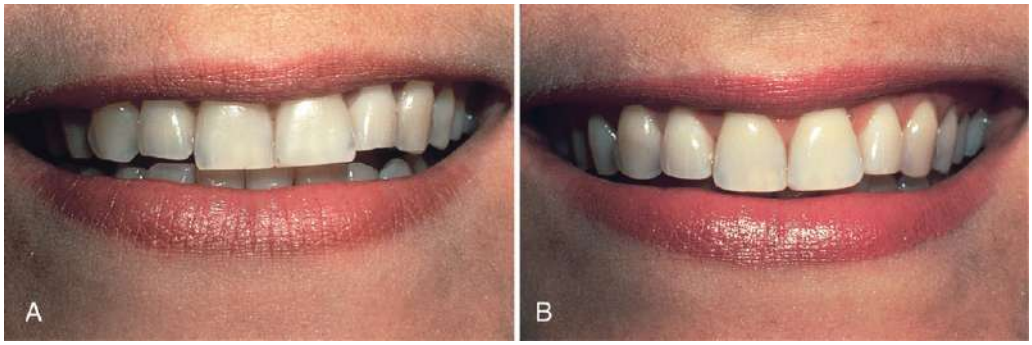


Figure 21.2 Cosmetic contouring. (A) Anterior teeth before treatment and (B) By reshaping teeth, a more youthful appearance is produced.

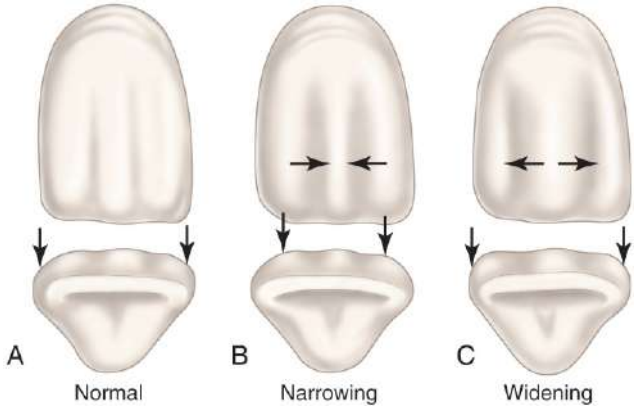


Figure 21.3 Creating illusions of width. (A) Normal width; (B) a tooth can be made to appear narrower by positioning mesial and distal line angles closer together and by more closely approximating developmental depressions and (C) greater apparent width is achieved by positioning line angles and developmental depressions farther apart.

- Compared with normal tooth contours (Fig. 21.3A), a tooth can be made to appear narrower by positioning the mesiofacial and distofacial line angles closer together (Fig. 21.3B).
- Developmental depressions also can be positioned closer together to enhance the illusion of narrowness.
- Similarly, greater apparent width can be achieved by positioning the line angles and developmental depressions further apart (Fig. 21.3C).

- Although more difficult, the apparent length of teeth also can be changed by illusion. Compared with normal tooth contours (Fig. 21.4A), a tooth can be made to appear shorter by
  - emphasizing the horizontal elements such as gingival perikymata
  - by positioning the gingival height of contour further incisally (Fig. 21.4B).
  - Slight modification of the incisal area, achieved by moving the incisal height of contour further gingivally, also enhances the illusion of a shorter tooth.
- The opposite tenets are true for increasing the apparent length of a tooth. The heights of contour are moved farther apart incisogingivally, and vertical elements such as developmental depressions are emphasized (Fig. 21.4C).

Used in combination, these illusionary techniques are particularly valuable for controlling the apparent dimension of teeth in procedures that result in an actual increased width of teeth, such as in diastema (i.e. space) closure (see section Correction of Diastemas). By contouring the composite additions in such a way that the original positions of the line angles are maintained, the increased widths of the restored teeth are less noticeable (Fig. 21.5). If full facial coverage restorations are placed in conjunction with a diastema closure, vertical elements can be enhanced and horizontal features deemphasized to control further the apparent dimension of teeth.

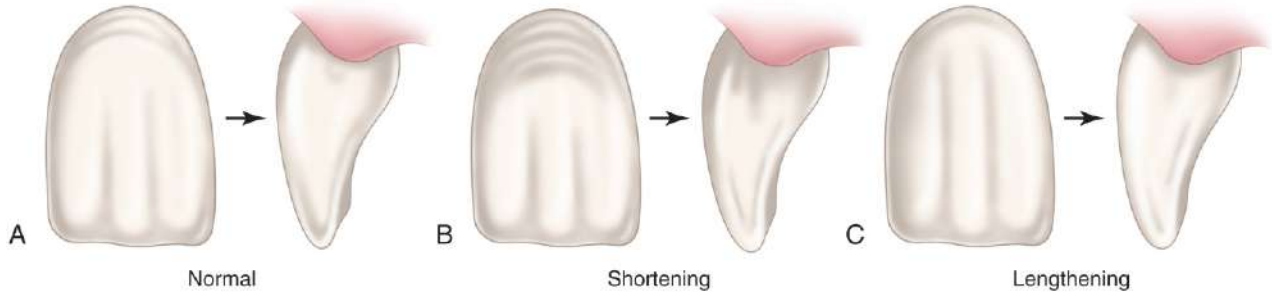


Figure 21.4 Creating illusions of length. (A) Normal length; (B) a tooth can be made to appear shorter by emphasizing horizontal elements and by positioning the gingival height of contour farther incisally and (C) the illusion of length is achieved by moving the gingival height of contour gingivally and by emphasizing vertical elements, such as developmental depressions.



Figure 21.5 Controlling apparent tooth size when adding proximal dimension. (A) Teeth before treatment and (B) by maintaining original positions of the facial line angles (see areas of light reflection), increased widths of teeth after composite augmentations are less noticeable.

Symmetry and Proportionality

Symmetry

The overall esthetic appearance of a human smile is governed largely by the symmetry and proportionality of the teeth that constitute the smile. Asymmetric teeth or teeth that are out of proportion to the surrounding teeth disrupt the sense of balance and harmony essential for optimal esthetics. Assuming that the patient's teeth are aligned normally (i.e. rotations or faciolingual positional defects are not present), *dental symmetry can be maintained if the sizes of contralateral teeth are equivalent*.

- A dental caliper should be used in conjunction with any conservative esthetic dental procedure that would alter the mesiodistal dimension of teeth. This recommendation is particularly true for procedures such as diastema closure or other procedures involving augmentation of proximal surfaces with composite. By first measuring and recording the widths of the interdental space and the teeth to be augmented, the appropriate amount of contour to be generated with composite resin addition can be determined (Fig. 21.6). In this manner, symmetric and equal tooth contours can be generated (see section Correction of Diastemas).
- When dealing with restorations involving the midline, particular attention also must be paid to the incisal and gingival embrasure forms; the mesial contours of both central incisors must be mirror images of one another to ensure an optimally symmetric and esthetic result.

Proportionality

In addition to being symmetric, anterior teeth must be in proper proportion to one another to achieve maximum esthetics. The quality of proportionality is relative and varies greatly, depending on other factors (e.g. tooth position, tooth alignment, arch form, configuration of the smile).

- One long-accepted theorem of the relative proportionality of maxillary anterior teeth typically visible in a smile involves the concept of *the golden proportion*.[2] Originally formulated as one of Euclid's elements, this theorem has been relied on through the ages as a geometric basis for proportionality in the beauty of art and nature.[3] On the basis of this formula, a smile, when viewed from the front, is considered to be esthetically pleasing *if each tooth in that smile (starting from the midline) is approximately 60% of the apparent mesiodistal width of the tooth immediately mesial to it*. The exact proportion of the distal tooth to the mesial tooth is **0.618** (Fig. 21.7A).
- These recurring esthetic dental proportions are based on the *apparent* mesiodistal width of teeth when viewed straight on and not the *actual* mesiodistal width of individual teeth. An example of an esthetically pleasing smile that meets the golden proportion can be seen in Fig. 21.7B.
- Although the golden proportion is not the absolute determinant of dental esthetics, it does provide a practical and proven guide for establishing proportionality when restoring anterior teeth, especially if these teeth are considered long. However, according to a study by Preston, only 17% of the population naturally exhibits smiles that meet the golden proportion as the recurring esthetic dental proportion.[4] According to a survey of dentists by Ward, a *recurring esthetic dental proportion of 70% (as opposed to 61.8% as with the golden proportion) is preferred when teeth are of normal dimension*.[5]

Considering the range of recommended width-to-length ratios[6–8] a good benchmark for achieving optimally esthetic results are

- *maxillary central incisors: 0.80*
- *maxillary lateral incisors: 0.75*

A good example can be seen in Fig. 21.7C–F. Because of altered passive eruption, this teenaged patient exhibited a preoperative width-to-length ratio of 1:1 for her maxillary central incisors (Fig. 21.7C). She revealed excessive gingival tissues relative to the display of her anterior teeth. To achieve a more esthetic appearance, it was determined that a width-to-length ratio of 0.80 would be desirable. After a periodontal surgical crown-lengthening procedure, a more esthetic width-to-length ratio of 0.80 was attained (Fig. 21.7D). After placement of eight porcelain veneers for



Figure 21.6 Diastema closure. (A) Teeth before composite additions and (B) symmetrical and equal contours are achieved in the final restorations.

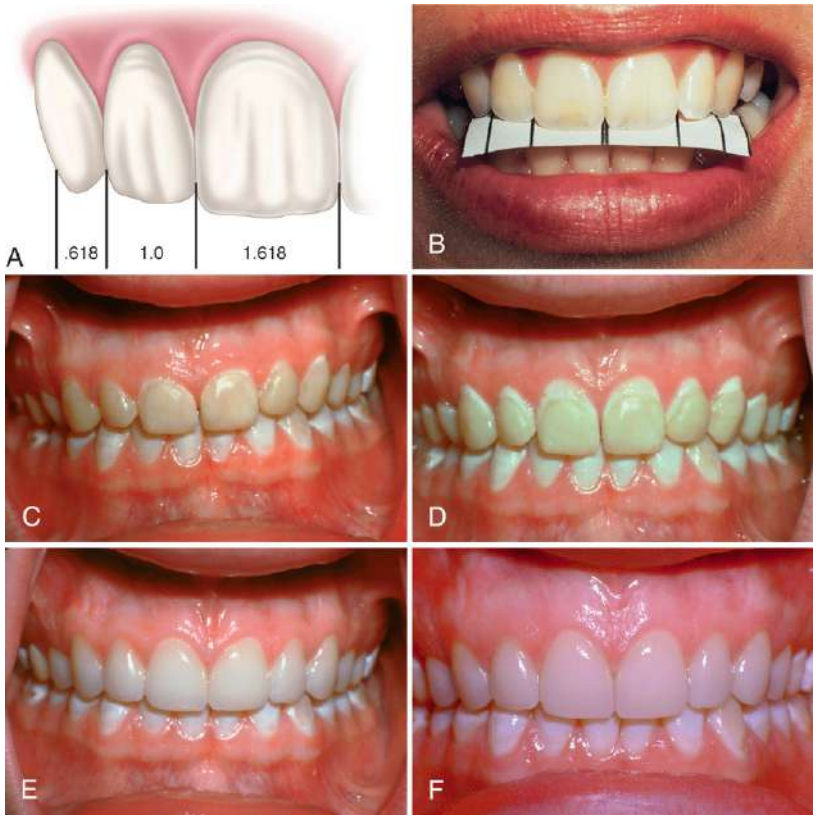


Figure 21.7 Tooth proportions. (A) The rule of the golden proportion. The exact ratios of proportionality. (B) The anterior teeth of this patient are in golden proportion to one another. (C) Width-to-length ratios. Preoperative view reveals width-to-length ratio of 1:1. (D) Following a periodontal surgical crown lengthening procedure, a more esthetic width-to-length ratio of 0.80 exists. (E) Final result with etched porcelain veneers. A width-to-length ratio of 0.80 is maintained. (F) A 7-year postoperative view reveals a stable and esthetic long-term result.

the treatment of fluorosis discolouration, the same 0.80 width-to-length ratio can be seen (Fig. 21.7E). A 7-year postoperative photograph reveals the long-term esthetic result (Fig. 21.7F).

Because central incisors are the dominant focal point in dental composition, the dentist must avoid narrow, elongated or short-and-wide contours. Adequate treatment planning and a fundamental knowledge of the importance of the ideal width-to-length ratios can optimize the final esthetic outcome. The importance of interdisciplinary treatment involving orthodontics, periodontics or both cannot be underestimated.

Position and Alignment

The overall harmony and balance of a smile depend largely on proper position of teeth and their alignment in the arch. Malposed or rotated teeth disrupt the arch form and may interfere with the apparent relative proportions of teeth. Orthodontic treatment of such defects always should be considered, especially if other positional or malocclusion problems exist in the mouth. If orthodontic treatment is either impractical or unaffordable, however, minor positional defects often can be treated with composite augmentation or full facial veneers indirectly made from composite or porcelain. Only problems that can be treated conservatively without significant alteration of the occlusion or gingival contours of teeth should be treated in this manner.

Minor rotations can be corrected by reducing the enamel in the area of prominence and augmenting the deficient area with composite resin (Fig. 21.8A and B). Care must be taken to restrict all recontouring of prominent areas to enamel. If the rotation is to be treated with an indirectly fabricated composite or porcelain veneer, an intra-enamel preparation is recommended, with greater reduction provided in the area of prominence. This preparation allows subsequent restoration to appropriate physiologic contours.

Malposed teeth are treated in a similar manner. Teeth in mild linguoversion can be treated by augmentation with full facial veneers placed directly with composite or made indirectly from processed composite or porcelain (Fig. 21.8C and D). Care must be exercised in maintaining physiologic gingival contours that do not impinge on tissue or result in an emergence profile of the restoration that is detrimental to gingival health. A functional incisal edge should be maintained by appropriate contouring of the restoration (an excessively thick incisal edge should be avoided). If the occlusion allows, limited reduction of enamel on the lingual aspect can be accomplished to

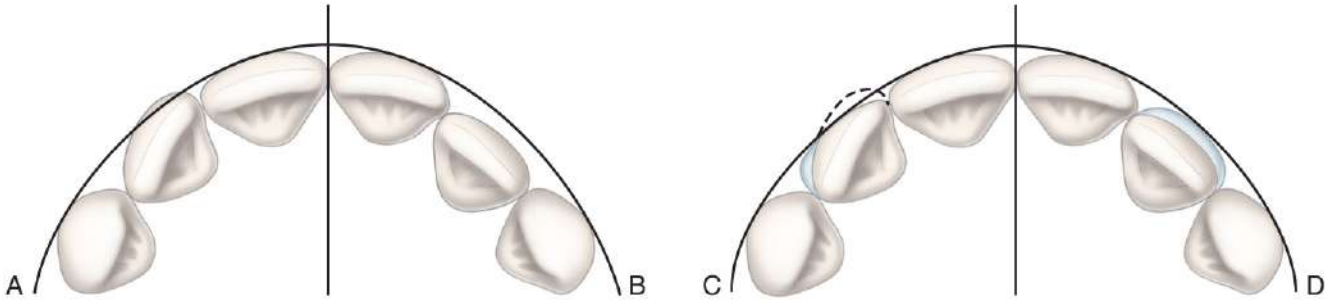


Figure 21.8 Position and alignment. (A) A minor rotation is first treated by reducing enamel in the area of prominence; (B) the deficient area is restored to proper contour with composite; (C) maxillary lateral incisor is in slight linguoversion and (D) restorative augmentation of facial surface corrects malposition.

reduce the faciolingual dimension of the incisal portion of the tooth. Individual teeth that are significantly displaced facially (i.e. facioversion) are best treated orthodontically.

Surface Texture

The character and individuality of teeth are determined largely by their surface texture and existing characteristics. Realistic restorations closely mimic the subtle areas of stippling, concavity and convexity that are typically present on natural teeth. Teeth in young individuals characteristically exhibit significant surface characterization, whereas teeth in older individuals tend to possess a smoother surface texture caused by abrasional wear. Even in older patients, however, restorations that are devoid of surface characterizations are rarely indicated.

The surfaces of natural teeth typically break up light and reflect it in many directions. Consequently, anatomic features (e.g. developmental depressions, prominences, facets, perikymata) should be examined closely and reproduced to the extent that they are present on the surrounding surfaces. The restored areas of teeth should reflect light as in the unrestored adjacent surfaces. In addition, as alluded to earlier, by controlling areas of light reflection and shadowing, various desired illusions also can be created.

Colour

Colour is the most complex and least understood artistic element. It is an area in which numerous interdependent factors exist, all of which contribute to the final esthetic outcome of the restoration. Although complex, a basic knowledge of colour is imperative to producing consistently esthetic restorations (Chapter 17).

Three fundamental elements of colour are *hue*, *value* and *chroma*:

- Hue is the intrinsic quality or shade of the colour.
- Value refers to the relative lightness or darkness of a hue. It is determined by the amount of white or black in a hue.
- Chroma is the intensity of any particular hue. Some current shade guides are based first on value because of the importance of this element of colour.

Dentists also must understand the colouration of natural teeth to select the appropriate shades of restorative materials accurately and consistently. Teeth typically are composed of a multitude of colours.

Clinical Considerations of Colour

- A *gradation of colour* usually occurs from the gingival region to the incisal region, with the gingival region being typically darker because of thinner enamel.
- Exposed *root surfaces* are particularly darker (i.e. dentin coloured) because of the absence of overlying enamel.
- In most individuals, *canines are slightly darker* than are incisors.
- *Young individuals* with thick enamel characteristically exhibit lighter teeth.
- Individuals with *darker complexions* usually appear to have lighter teeth because of the contrast that exists between teeth and the surrounding facial structures.
- Women can enhance the apparent lightness of their teeth simply by using a darker shade of makeup or lipstick. By increasing the contrast between teeth and the surrounding facial tissue, the illusion of lighter teeth can be created.
- Colour changes associated with ageing also occur, primarily owing to wear. As the facial enamel is worn away, the underlying dentin becomes more apparent, resulting in a darker tooth.
- Incisal edges are often darker because of the thinning of enamel or the exposure of dentin because of normal attrition. Cervical areas also tend to darken because of abrasion.

An understanding of normal tooth colouration enhances the dentist's ability to create a restoration that appears natural. Several clinical factors also must be considered, however, to enhance the colour-matching quality of the restoration. Many shade guides for composite materials are inaccurate. Not only are they often composed of a material dissimilar to that of the composite, but they also do not take into consideration colour changes that occur from batch to batch or changes caused by ageing of the composite.

- Accurate shade selection is best attained by applying and curing a small amount of the composite restorative material in the area of the tooth that may need restoration.
- Shade selection should be determined before isolating teeth so that colour variations that can occur as a result of drying and dehydration of teeth are avoided.
- Problems in colour perception also complicate selection of the appropriate shade of restorative material. Various light sources produce different perceptions of colour. This phenomenon is referred to as *metamerism*.[9] The colour of the surrounding environment influences what is seen in the patient's mouth. Colour perception also is influenced by the physiologic limitations of the dentist's eye. On extended viewing of a particular tooth site, eyes experience colour fatigue, resulting in a loss of sensitivity to yellow-orange shades.[7] By looking away at a blue object or background (i.e. the complementary colour), the dentist's eyes quickly recover and are able to distinguish subtle variations in yellow-orange hues again.

Because of the many indirect factors that influence colour perception, it is recommended that the dentist, the assistant and especially the patient all be involved in shade selection.

Translucency

Translucency is another factor that affects the esthetic quality of the restoration. The degree of translucency is related to how deeply light penetrates into the tooth or restoration before it is reflected outwards. Normally, light penetrates through enamel into dentin before being reflected outwards (Fig. 21.9A); this affords the realistic esthetic vitality characteristic of normal, unrestored teeth. Shallow penetration of light often results in loss of esthetic vitality. This phenomenon is a common problem encountered when treating severely intrinsically stained teeth such as teeth affected by tetracycline with direct or indirect veneers. Although opaque resin media can mask the underlying stain, esthetic vitality is usually lost because of reduced light penetration (Fig. 21.9B). Indirect veneers of processed composite or porcelain fabricated to include inherent opacity also may have this problem.

Illusions of translucency also can be created to enhance the realism of a restoration. Colour modifiers (also referred to as *tints*) can be used to achieve apparent translucency and tone down bright stains or to characterize a restoration. Fig. 21.10 shows a case in which the maxillary right central incisor with intrinsic yellow staining caused by trauma to the tooth warranted restoration. When bleaching treatments were unsuccessful because of calcific

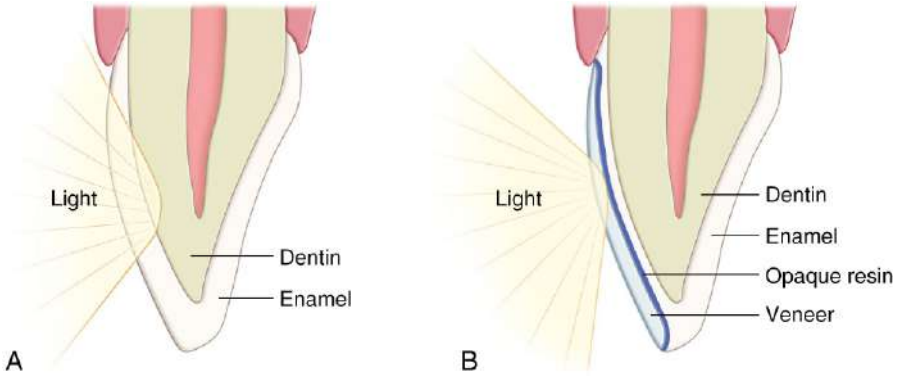


Figure 21.9 Translucency and light penetration. (A) Light normally penetrates deeply through enamel and into dentin before being reflected outwards. This affords realistic esthetic vitality. (B) Light penetration is limited by opaquing resin media under veneers. Esthetic vitality is compromised.



Figure 21.10 Use of internally placed colour modifiers. (A) The maxillary right central incisor exhibits bright intrinsic yellow staining as a result of calcific metamorphosis and (B) Colour modifiers under direct-composite veneer reduce brightness and intensity of stain and simulate vertical areas of translucency.

metamorphosis, a direct composite veneer was used in this patient. After an intra-enamel preparation and acid etching, a violet colour modifier (the complementary colour of yellow) was applied to the prepared facial surface to reduce the brightness and intensity of the underlying yellow tooth. Additionally, a mixture of grey and violet colour modifiers was used to simulate vertical areas of translucency. The final restoration is shown in Fig. 21.10B. Colour modifiers also can be incorporated in the restoration to simulate maverick colours, to check lines or to surface spots for further characterization. Colour modifiers always should be placed within a composite restoration, not on its surface.

Clinical Considerations

Although an understanding of basic artistic elements is imperative to successfully placing esthetic restorations, certain clinical considerations must be addressed concomitantly to ensure the overall quality of the restoration. In addition to being esthetic, restorations must be functional. Dawson stated that '*Esthetics and function go hand in hand. The better the esthetics, the better the function is likely to be and vice versa.*'[10]

- The occlusion always must be assessed before any conservative esthetic procedure.
- Anterior guidance, in particular, must be maintained and occlusal harmony ensured when treating areas involved in occlusion.
- Another requirement of all conservative esthetic restorations is that they possess physiologic contours that promote good gingival health. Particular care must be taken in all treatments to finish the gingival areas of the restoration adequately and to remove any gingival excess of material.
- Emergence angles of the restorations must be physiologic and not impinge on gingival tissue.

Conservative Alterations of Tooth Contours and Contacts

Many unsightly tooth contours and diastemas can be corrected or the appearance greatly improved by several conservative methods. Often these procedures can be incorporated into routine restorative treatment. The objective is to improve esthetics and yet preserve as much healthy tooth structure as possible, consistent with the acceptable occlusion and health of surrounding tissue. These procedures include reshaping natural teeth, correcting embrasures and closing diastemas.

Alterations of Shape of Natural Teeth

Some esthetic problems can be corrected conservatively without the need for tooth preparation and restoration. Consideration always should be given to reshaping and polishing natural teeth to improve their appearance and function (Fig. 21.11). In addition, the rounding of sharp angles can be considered a prophylactic measure to reduce stress and to prevent chipping and fractures of the incisal edges.

Etiology

Attrition of the incisal edges often results in closed incisal embrasures and angular incisal edges (Fig. 21.11A). Anterior teeth, especially maxillary central incisors, often are fractured in accidents. Other esthetic problems, including

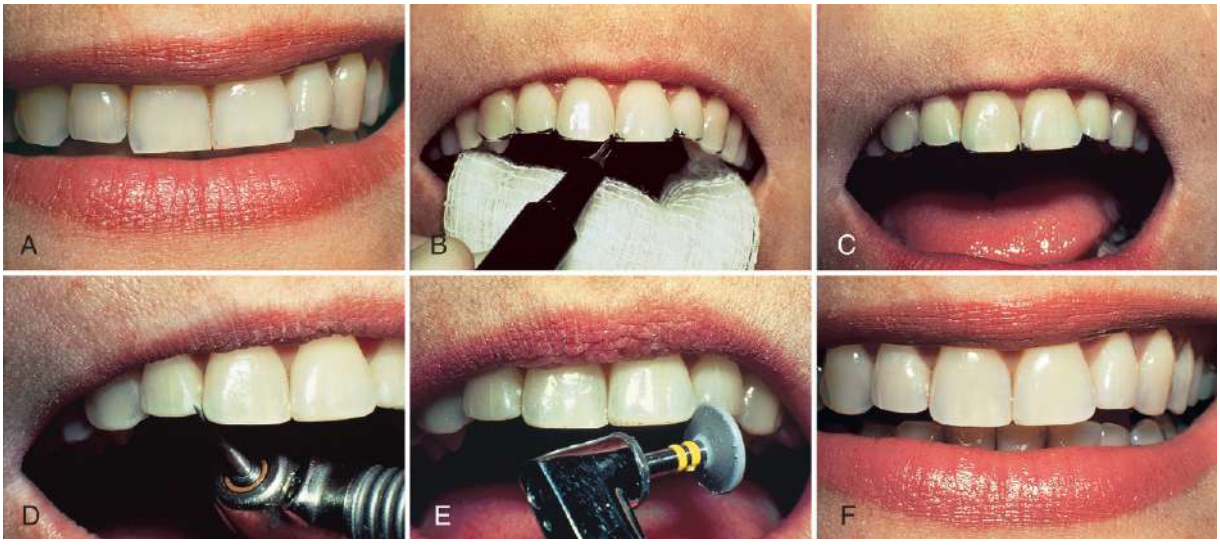


Figure 21.11 Reshaping natural teeth. (A) Maxillary anterior teeth with worn incisal edges; (B) areas to be reshaped are outlined; (C) outlined areas give the patient an idea of what the final result will look like; (D) a diamond instrument is used to reshape the incisal edges; (E) a rubber abrasive disk is used to polish the incisal edges and (F) reshaping results in a more youthful smile.

attrition and abnormal wear from poor dental habits (e.g. biting fingernails, holding objects with teeth), often can be corrected or the appearance improved by reshaping natural teeth.

Treatment

- Consultation and examination are necessary before any changes are made to the shapes of teeth. Photographs, study models, line drawings and esthetic imaging devices enable the patient to envision the potential improvement before any changes are made.
- As noted earlier, cosmetic contouring to achieve youthful characteristics often includes rounding incisal angles, reducing facial line angles and opening incisal embrasures. The opposite characteristics typically are considered more mature features. Cosmetic reshaping to smooth rough incisal edges and improve symmetry is equally beneficial to women and men.
- The patient must understand what is involved and must want to have the alteration made. If reshaping is desired, it is helpful to mark, by using a pencil or alcohol-marking pen, an outline of the areas of the teeth to be reshaped (Fig. 21.11B). By marking the anticipated areas for enamel reshaping, the patient is provided some indication of what the postoperative result may look like (Fig. 21.11C). If available, esthetic computer imaging also can be used to illustrate the possible result before treatment.
- Because all reshaping is restricted to enamel, anaesthesia is not required. A cotton roll is recommended for isolation. Diamond instruments and abrasive disks and points are used for contouring, finishing and polishing (Fig. 21.11D and E). Through careful reshaping of appropriate enamel surfaces, a more esthetic smile (characterized by youthful features) is attained. Rounded incisal edges also are less likely to chip or fracture (Fig. 21.11F).
- A second example involves the irregular, fractured incisal surfaces of maxillary central incisors (Fig. 21.12A). An esthetic result can be accomplished by slightly shortening the incisal edges and reshaping both teeth to a symmetrical form. Photographs, line drawings, esthetic imaging or marking the outline on the patient's teeth enables the patient to envision the potential improvement before any changes are made. Protrusive function always should be evaluated to prevent inadvertent elimination of this occlusal contact. Conservative treatment consists of using diamond instruments and abrasive disks and points for contouring and polishing the central incisors. The finished result is illustrated in Fig. 21.12B.
- As some patients grow older or have the habit of bruxism, the incisal surfaces often wear away, leaving sharp edges that chip easily. This is also accompanied by loss of the incisal embrasures (Fig. 21.13A). To lessen the chance of more fractures and to create a more youthful smile, the incisal embrasures are opened and the incisal angles of teeth are rounded (Fig. 21.13B).



Figure 21.12 Irregular incisal edges. (A) Central incisors have rough, fractured incisal edges and (B) esthetic result is obtained by recontouring incisal edges.

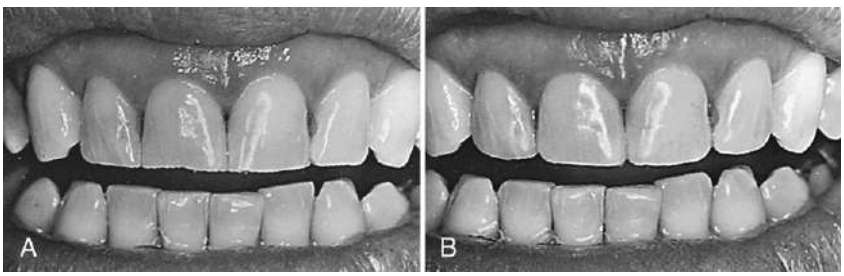


Figure 21.13 Loss of incisal embrasures from attrition. Before (A) and after (B) recontouring teeth to produce a more youthful appearance and improve resistance to fracture.



Figure 21.14 Closing incisal embrasures. (A) Maxillary canines moved to close spaces left by missing lateral incisors. The mesial incisal embrasures are too open. (B) Canines reshaped to appear like lateral incisors.

Alterations of Embrasures

Etiology

Anterior teeth can have embrasures that are too open as a result of the shape or position of teeth in the arch. When the permanent lateral incisors are congenitally missing, canines and posterior teeth may drift mesially or the space may be closed orthodontically. The facial surface and cusp angle of some canines can be reshaped to appear like lateral incisors. In many instances, the mesio-incisal embrasures remain too open (Fig. 21.14A).

Treatment

Composite can be added to establish an esthetic contour and correct the open embrasures. Evaluation of the occlusion before restoration determines if the addition would be compatible with functional movements. The patient should understand the procedures involved and should want to have the change made. Line drawings, esthetic imaging or photographs of similar examples are often helpful in explaining the procedure and allaying patient concerns. Another patient aid involves adding composite to teeth (unetched/unbonded) to fill the embrasure temporarily to simulate the final result, which is known as a *composite mockup*.

Preliminary procedures include cleaning the involved teeth, selecting the shade and isolating the area. Local anaesthesia usually is not required because the preparation does not extend sub-gingivally and involves only enamel. A coarse, flame-shaped diamond instrument is used to remove overly convex enamel surfaces (if present) and to roughen the enamel surface area to be augmented with composite material. It may be necessary to place a wedge and use an abrasive strip to prepare the proximal surface. The final contour of the restoration should be envisioned before the preparation is made so that all areas to be bonded are adequately roughened.

A polyester strip or a Teflon tape is inserted to protect the adjacent tooth during acid etching. After etching, rinsing and drying, the contoured strip is positioned. A light-cured composite material is inserted, and the strip is closed during polymerization. The incisal embrasures of both canines are corrected, and both restorations are finished by routine procedures (Fig. 21.14B). The occlusion should be evaluated to assess centric contacts and functional movements, and any adjustments or corrections should be made, if indicated.

Correction of Diastemas

Etiology

The presence of diastemas between anterior teeth is an esthetic problem for some patients (Fig. 21.15A). Before treatment, a diagnosis of the cause is made, including an evaluation of the occlusion. A frequent site of a diastema is between maxillary central incisors. Diastemas should not be closed without first recognizing and treating the underlying cause, as merely treating the cause may correct the diastema.

- A prominent labial frenum with nonelastic fibres extending proximally often prevents the normal approximation of erupting central incisors.[11]
- Other causative factors include
  - congenitally missing teeth,
  - undersized or malformed teeth,
  - inter-arch tooth size discrepancies (i.e. Bolton discrepancy),
  - supernumerary teeth and
  - heredity.
- Diastemas also may result from other problems such as tongue thrusting, periodontal disease or posterior bite collapse.

Treatment

Traditionally, diastemas have been treated by surgical, periodontal, orthodontic and prosthetic procedures. These types of corrections can be impractical or unaffordable and often do not result in permanent closure of the diastema. In carefully selected cases, a more practical alternative is use composite augmentation of proximal surfaces via adhesive procedures (Figs. 21.5 and 21.6). All treatment options (including no treatment) should be considered before resorting immediately to composite augmentation. Line drawings, photographs, computer



Figure 21.15 Diastema closure. (A) Esthetic problem created by space between central incisors; (B and C) interdental space and size of central incisors measured with caliper; (D) teeth isolated with cotton rolls and retraction cord tucked into the gingival crevice; (E) a diamond instrument is used to roughen enamel surfaces; (F) etched enamel surface indicated (*arrow*). (G) composite inserted with composite instrument; (H) matrix strip closed with thumb and forefinger; (I) composite addition is cured; (J) finishing strip used to finalize contour of first addition; (K) a tight contact is attained by displacing the second tooth being restored in a distal direction with thumb and forefinger, while holding matrix in contact with adjacent restoration; (L) flame-shaped finishing bur used to contour restoration; (M) finishing strip used to smooth the sub-gingival areas; (N) the restoration is polished with a rubber abrasive point; (O) final lustre attained with polishing paste applied with prophy cup; (P) unwaxed floss used to detect any excess composite and (Q) diastema closed with symmetrical and equal additions of composite.



Figure 21.16 Multiple diastemas occurring among maxillary anterior teeth. (A) Before correction and (B) appearance after diastemas are closed with composite augmentation.

imaging, models with spaces filled and direct temporary additions of ivory-coloured wax or composite material on natural teeth (unetched) are important preliminary procedures.

The correction of a diastema between maxillary central incisors is described and illustrated in Fig. 21.15.

- After the teeth are cleaned and the shade selected, a Boley gauge or other suitable caliper is used to *measure the width of the diastema and the individual teeth* (Fig. 21.15B and C). Occasionally, one central incisor is wider, requiring a greater addition to the narrower tooth. Assuming the incisors are of equal width, symmetrical additions can be ensured by using half of the total measurement of the diastema to gauge the width of the first tooth restored. Cotton rolls, instead of a rubber dam, are recommended for isolation because of the importance of relating the contour of the restoration directly to the proximal tissue. Usually, the restoration must begin slightly below the gingival crest to appear natural and to be confluent with the tooth contours.
- With cotton rolls in place, a *gingival retraction cord* of an appropriate size is tucked in the gingival crevice of each tooth from midfacial mesially to midlingual (Fig. 21.15D). The cord retracts the soft tissue and prevents seepage from the crevice. In some instances, the retraction cord may need to be inserted for one tooth at a time to prevent strangulation of interproximal tissue during preparation and restorative procedures.
- To enhance retention of the composite, a *coarse, flame-shaped diamond instrument* is used to roughen the proximal surfaces, extending from the facial line angle to the lingual line angle (Fig. 21.15E). More extension may be needed to correct the facial or lingual contours, depending on the anatomy and position of the individual tooth. The enamel is acid etched approximately 0.5 mm past the prepared, roughened surface. The acid should be applied gingivally only to the extent of the anticipated restoration. After rinsing and drying, the etched enamel should display a lightly frosted appearance (Fig. 21.15F). A 5-cm-by-5-cm (2-in-by-2-in) gauze is draped across the mouth and tongue to prevent inadvertent contamination of the etched preparations by the patient. After both preparations are completed, the teeth are restored one at a time.
- A polyester strip is contoured and placed proximally, with the gingival aspect of the strip extending below the gingival crest. Additional contouring may be required to produce enough convexity in the strip. In most cases, a wedge cannot be used.
- The strip is held (with the index finger) on the lingual aspect of the tooth to be restored, while the facial end is reflected for access. A light-cured composite is used for the restoration. After the bonding agent is applied, the composite material is inserted with a hand instrument (Fig. 21.15G).
- Careful attention is given to pressing the material lingually to ensure confluence with the lingual surface. The matrix is gently closed facially, beginning with the gingival aspect (Fig. 21.15H). Care must be taken not to pull the strip too tightly because the resulting restoration may be under-contoured faciolingually, mesiodistally or both.
- The light-cured composite material is polymerized with the light directed from the facial and lingual directions for an appropriate amount of time. Note that curing times may vary according to the type of light source, the composite used and the thickness of the material (Fig. 21.15I). Initially, it is better to over-contour the first restoration slightly to facilitate finishing it to an ideal contour.
- When polymerization is complete, the strip is removed. Contouring and finishing are achieved with appropriate carbide finishing burs, fine diamonds or abrasive disks (Fig. 21.15L). Finishing strips are invaluable for finalizing the proximal contours (Fig. 21.15J and M). When composite is being added to two adjoining surfaces, which is often the case, final polishing is deferred until the contralateral restoration is completed. However, it is sometimes necessary to utilize finishing discs to contour the first composite addition to achieve the correct mesiodistal dimension of the first tooth before the second tooth is restored. It is imperative for good gingival health that the cervical aspect of the composite addition be immaculately smooth and continuous

with the tooth structure. Overhangs must not be present. Removal of the gingival retraction cord facilitates inspection and smoothing of this area.

- Flossing with a length of unwaxed floss verifies that the gingival margin is correct and smooth if no fraying of the floss occurs (Fig. 21.15P).
- After etching, rinsing and drying, the second restoration is completed. A tight proximal contact can be attained by displacing the second tooth being restored in a distal direction (with the thumb and the index finger) while holding the matrix in contact with the adjacent restoration (Fig. 21.15K).
- Contouring is accomplished with a 12-fluted carbide bur and finishing strips (Fig. 21.15L and M). Articulating paper should be used to evaluate the patient's occlusion to ensure that the restorations are not offensive in centric or functional movements; adjustments can be made with a carbide finishing bur or abrasive disks.
- Final polishing is achieved with rubber polishing points or polishing paste applied with a prophy cup in a low-speed handpiece (Fig. 21.15N and O). Unwaxed floss is used to detect any excess material or overhang (Fig. 21.15P). The final esthetic result is seen in Fig. 21.15Q.

Clinical Scenarios

I. Multiple diastemas among the maxillary anterior teeth are shown in Fig. 21.16A. Closing the spaces by orthodontic movement was considered in this patient; however, because the patient's teeth were under-contoured mesio-distally, the diastemas were closed by bonding composite to the proximal surfaces. The teeth after treatment are shown in Fig. 21.16B.

II. In the presence of defective Class III restorations or proximal caries, it is recommended that the teeth be restored with the same composite used for closing the diastema. Often these restorations can be performed at the same time the diastema is closed with composite additions (Fig. 21.17).

III. Occasionally, diastemas are simply too large to close esthetically with composite augmentation alone (Fig. 21.18A). Closing a large space of this magnitude with composite would merely create an alternative esthetic problem—that is, excessively large central inci-

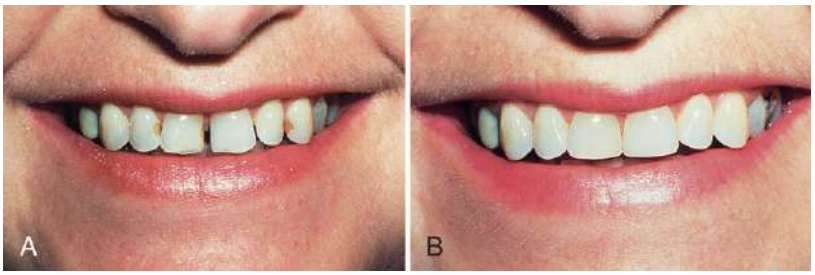


Figure 21.17 (A) Diastema closure and cosmetic contouring and (B) Significant esthetic improvement is achieved by replacing defective Class III restorations and closing diastemas with conservative composite additions and cosmetically reshaping teeth.

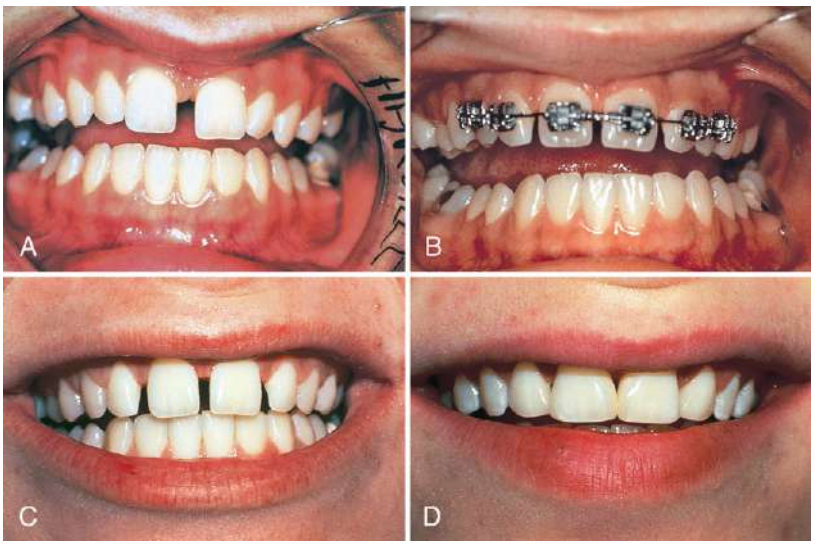


Figure 21.18 Space distribution. (A) Midline diastema too large for simple closure with composite additions; (B and C) space distributed among four incisors with orthodontic treatment and (D) final result after composite additions.

sors, which would further exacerbate the existing discrepancy in proportionality among anterior teeth. In such cases, large spaces are best redistributed orthodontically among anterior teeth so that symmetric and equal composite additions can be made to the central and lateral incisors (Fig. 21.18B and C). This approach that involves space distribution results in improved proportionality among anterior teeth (see section Artistic Elements). The final result, immediately after completion, is shown in Fig. 21.18D.

Conservative Treatments for Discoloured Teeth

One of the most frequent reasons patients seek dental care is discoloured anterior teeth. Even patients with teeth of normal colour request whitening procedures. Treatment options include removal of surface stains, bleaching, micro-abrasion or macro-abrasion, veneering and placement of porcelain crowns. Many dentists recommend porcelain crowns as the best solution for badly discoloured teeth. If crowns are done properly with the highly esthetic ceramic materials currently available, they have great potential for being esthetic and long lasting. However, increasing numbers of patients do not want their teeth 'cut down' for crowns and choose an alternative, conservative approach such as veneers (section Veneers) that preserves as much of the natural tooth as possible. This treatment is performed with the understanding that the corrective measures may be less permanent.

Discolourations are classified as extrinsic or intrinsic. Extrinsic stains are located on the outer surfaces of teeth, whereas intrinsic stains are internal. The etiology and treatment of extrinsic and intrinsic stains are discussed in the following sections.

I. Extrinsic Discolourations

Etiology

Stains on the external surfaces of teeth (referred to as *extrinsic discolourations*) are common and may be the result of numerous factors.



Figure 21.19 Extrinsic surface stains on facial surfaces in a young patient.

- In young patients, stains of almost any colour can be found and are usually more prominent in the cervical areas of teeth (Fig. 21.19). These stains may be related to remnants of Nasmyth membrane, poor oral hygiene, existing restorations, gingival bleeding, plaque accumulation, eating habits or the presence of chromogenic microorganisms.[12]
- In older patients, stains on the surfaces of teeth are more likely to be brown, black or grey and occur on areas adjacent to gingival tissue. Poor oral hygiene is a contributing factor, but coffee, tea and other types of chromogenic foods or medications can produce stains (even on plaque-free surfaces).
- Tobacco stains also are observed frequently. Existing restorations may be discoloured for the same reasons.

Treatment

Most surface stains can be removed by routine prophylactic procedures (Fig. 21.20); however, some superficial discolourations on tooth-coloured restorations and decalcified areas on teeth cannot be corrected by such cleaning. Conservative correction may be accomplished by mild microabrasion or by surfacing the thin, outer, discoloured layer with a flame-shaped, carbide finishing bur or diamond instrument (i.e. macro-abrasion), followed by polishing with abrasive disks or points to obtain an acceptable



Figure 21.20 Treatment of surface stains. (A) Tobacco stains; (B) pumicing teeth with rubber cup and (C) shade guide used to confirm normal colour of natural teeth.

result. (See section Micro-abrasion and Macro-abrasion for details of clinical technique.)

II. Intrinsic Discolourations

Etiology

Intrinsic discolourations are caused by deeper (not superficial) internal stains or enamel defects and are more complex to treat than are external discolourations. Teeth with vital or nonvital pulps as well as endodontically treated teeth can be affected.

Vital teeth discolouration

The staining may be located in enamel or dentin. Discolourations restricted to dentin still may show through enamel. Discolouration also may be localized or generalized, involving the entire tooth.

i. During tooth development

Vital teeth may be discoloured at the time the crowns are forming and the abnormal condition usually involves several teeth. Causative factors include

- hereditary disorders,
- medications (particularly tetracycline preparations),
- excess fluoride,
- high fevers associated with early childhood illnesses and
- other types of trauma.[12]

Localized areas of discolouration may occur on individual teeth because of enamel or dentin defects induced during tooth development. High fevers and other forms of trauma can damage the tooth during its development, resulting in unesthetic hypoplastic defects. Additionally, localized areas of dysmineralization, or the failure of the enamel to calcify properly, can result in hypocalcified white spots.

ii. Tertracycline staining

Various preparations of the antibiotic drug tetracycline can cause the most distracting, generalized type of intrinsic discolouration (Fig. 21.21A).[13] The severity of the staining depends on the dose, the duration of exposure to the drug and the type of tetracycline analogue used. Different types of tetracyclines induce different types of discolouration, varying from yellow-orange to dark blue-grey. Dark blue-grey, tetracycline-stained teeth are considerably more difficult to treat than are teeth with mild yellow-orange discolourations. Staining from tetracycline-type drugs most frequently occurs at an early age and is caused by ingestion of the drug concomitant with the development of permanent teeth. Studies indicate that permanent teeth in adults also can experience a greying discolouration, however, as a result of long-term exposure to minocycline, a tetracycline analogue.[13]

iii. Fluorosis

The presence of excessive fluoride in drinking water and other sources at the time of teeth formation can result in another type of intrinsic stain called *fluorosis*. The staining usually is generalized.

iv. Poor oral hygiene

After eruption, poor oral hygiene also can result in decalcified white spots. Poor oral hygiene during orthodontic treatment frequently results in these types of decalcified defects. White or discoloured spots with intact enamel surface (i.e. surface not soft) are often evidence of intra-oral remineralization, however, and such spots are not indications for invasive treatment (unless for esthetic concerns).

v. Ageing

As noted earlier, ageing effects also can result in yellowed teeth. As patients grow older, the tooth enamel becomes thinner because of wear and allows underlying dentin to become more apparent. Also, often continuing deposition of secondary dentin occurs in older individuals, resulting in greater dentin thickness. This deposition results in a yellowing effect, depending on the intrinsic colour of dentin. Additionally, the permeability of teeth usually allows the infusion (over time) of significant organic pigments (from chromogenic foods, drinks and tobacco products) that produce a yellowing effect.

Additionally, caries, metallic restorations, corroded pins and leakage or secondary caries around existing restorations can result in various types of intrinsic discolouration.

Nonvital teeth discolouration

Nonvital teeth also can become discoloured intrinsically. These stains usually occur in individual teeth after eruption has taken place.



Figure 21.21 Intrinsic stains. (A) Staining by tetracycline drugs and (B) staining of the maxillary left central incisor from tooth trauma and degeneration of the pulp.



Figure 21.22 Illusion of a lighter appearance of teeth by use of darker makeup. (A) Before and (B) after. (From Freedman G: *Contemporary esthetic dentistry*, St. Louis, 2012, Mosby.)

I. The pulp may become infected or degenerate as a result of trauma, deep caries or irritation from restorative procedures. If these teeth are properly treated by endodontic therapy, they usually retain their normal colour. If treatment is delayed, discolouration of the crown is more likely to occur. The degenerative products from the pulp tissue stain dentin, and this is readily apparent because of the translucency of enamel (Fig. 21.21B).
II. Trauma resulting in *calcific metamorphosis* (i.e. calcification of the pulp chamber, root canal or both) also can produce significant yellowing of the tooth. This condition is extremely difficult to treat (Fig. 21.10).

Treatment

Some people definitely have esthetic problems because of intrinsic stains, but others worry needlessly about the overall colour of their teeth. In the latter instance, the dentist must decide if the colour of teeth can be improved enough to justify treatment, even though the patient insists on having something done. Individuals with light complexions may believe that their teeth are too dark, when actually they are normal in colour (Fig. 21.22A). Positioning a shade tab from a shade guide of tooth colours next to such teeth often shows these patients that the colour of their teeth is well within the normal range of shades. As stated earlier, tanned skin, darker makeup or darker lipstick usually make teeth appear much whiter by increasing the contrast between teeth and the surrounding facial features (Fig. 21.22B).

The patient should be told that many discolourations can be corrected or the appearance of teeth greatly improved through conservative methods such as bleaching, micro-abrasion or macro-abrasion and veneering. *Mild discolourations are best left untreated, are bleached or are treated conservatively with micro-abrasion or macro-abrasion because no restorative material is as good as the healthy, natural tooth structure*. The patient should be informed that the gingival tissue adjacent to restorative material will never be as healthy as that next to normal tooth structure.

Colour photographs of previously treated teeth with intrinsic staining (i.e. before and after treatment) are excellent adjuncts to help the patient make an informed decision. Esthetic imaging with modern computer simulation of the postoperative result also can be an effective educational tool. Patients appreciate knowing what the cause of the problem is, how it can be corrected, how much time is involved and what the cost will be. They also should be informed of the life expectancy of the various treatment alternatives suggested. Vital bleaching usually results in tooth lightening for only 1–3 years, whereas an etched porcelain veneer should last 10–15 years or longer. With continuous improvements in materials and techniques, a much longer lifespan may be possible with any of these procedures. The clinical longevity of esthetic restorations also is enhanced in patients with good oral hygiene, proper diet, a favourable bite relationship and little or no contact with agents that cause discolouration or deterioration.

Correction of intrinsic discolourations caused by failing restorations entails replacement of the faulty portion or the entire restoration. Correction of discolourations caused by caries lesions requires appropriate restorative treatment. Esthetic inserts for metal restorations are described later in this chapter. For the other types of intrinsic discolourations previously discussed, detailed treatment options are presented in the following three sections.

Bleaching Treatments

The lightening of the colour of a tooth through the application of a chemical agent to oxidize the organic pigmentation in the tooth is referred to as ***bleaching***.

In keeping with the overall conservative philosophy of tooth restoration, consideration should be given first to bleaching anterior teeth when intrinsic discolourations are encountered. Bleaching of nonvital teeth was first reported in 1848; in-office bleaching of vital teeth was first reported in 1868.[14] By the early 1900s, in-office vital bleaching had evolved to include the use of heat and light for activation

Box 21.1 Classification of bleaching treatments

I. **Nonvital bleaching procedures:**
   i. In-office nonvital bleaching technique
   ii. Walking bleach technique
II. **Vital bleaching procedures:**
   i. In-office vital bleaching technique
   ii. Dentist-prescribed, home-applied technique

of the process. Although a 3% ether and peroxide mouthwash used for bleaching in 1893 has been reported in the literature, the 'dentist-prescribed, home-applied' technique (also referred to as *nightguard vital bleaching* or *at-home bleaching*) for bleaching vital teeth outside the office began around 1968, although it was not commonly known until the late 1980s.[15]

The classification of the various types of bleaching treatments available are given in Box 21.1.

Mechanism of Bleaching

Most bleaching techniques use some form or derivative of hydrogen peroxide in different concentrations and application techniques. The mechanism of action of bleaching teeth with hydrogen peroxide is considered to be oxidation of organic pigments, although the chemistry is not well understood.

Clinical Note

- Bleaching generally has an approximate lifespan of 1–3 years, although the change may be permanent in some situations.
- With all bleaching techniques, a transitory decrease occurs in the potential bond strength of composite when it is applied to bleached enamel and dentin. This reduction in bond strength results from residual oxygen or peroxide residue in the tooth that inhibits the setting of the bonding resin, precluding adequate resin tag formation in the etched enamel.
- No loss of bond strength is noted if the composite restorative treatment is delayed at least 1 week after cessation of any bleaching.[16]

I. Nonvital Bleaching Procedures

The primary indication for nonvital bleaching is to lighten teeth that have undergone endodontic therapy. Discolouration may be a result of bleeding into dentin from trauma before root canal therapy, degradation of pulp tissue left in the chamber after such therapy or staining from restorative materials and cements placed in the tooth as a part of the endodontic treatment. Most posterior teeth that have received endodontic therapy require full-coverage restorations that encompass the tooth to prevent subsequent fracture. Anterior teeth needing restorative treatment and that are largely intact may be restored with composite rather than with partial-coverage or full-coverage restorations without significantly compromising the strength of the tooth.[17] This knowledge has resulted in a resurgence in the use of nonvital bleaching techniques.

Nonvital bleaching techniques include an in-office technique and an out-of-office procedure referred to as *walking bleach*. (See the following sections for details of these two techniques.)

A. In-Office Nonvital Bleaching Technique

Concept

- The in-office bleaching for nonvital teeth historically has involved a thermocatalytic technique consisting of the placement of *35% hydrogen peroxide liquid* into the debrided pulp chamber and acceleration of the oxidation process by placement of a heating instrument into the pulp chamber. The *thermocatalytic technique is not recommended*, however, because of the potential for cervical resorption (Fig. 21.23).[18]
- A more current technique *uses 30%–35% hydrogen peroxide pastes or gels that require no heat*. This technique is frequently the preferred in-office technique for bleaching nonvital teeth.
  - In both techniques, it is imperative that a *sealing cement* (resin-modified glass ionomer [RMGI] cement is recommended) be placed over the exposed endodontic filling before application of the bleaching agent to prevent leakage and penetration of the bleaching material in an apical direction.
  - It is also recommended that the bleaching agent be applied in the coronal portion of the tooth incisal to the level of the periodontal ligament (not down into the root canal space) to prevent unwanted leakage of the bleaching agent through the lateral canals or canaliculi to the periodontal ligament.



Figure 21.23 Radiograph revealing the presence of extensive cervical resorption.



Figure 21.24 Indication for bleaching root canal-filled tooth. (A) Before and (B) after intra-coronal, nonvital bleaching.

B. Walking Bleach Technique

Concept

The walking bleach technique is an in-office bleaching procedure that does not require the use of heat and employs *sodium perborate* as the bleaching agent. The bleaching process is slower and continues till the patient reports back for the subsequent appointment for assessment and cessation of the treatment (hence the name 'walking bleach') (Fig. 21.24).

Clinical Procedure

Step 1: isolation and access refinement

Before beginning the walking bleach technique, the dentist needs to evaluate the potential for occlusal contact on the area of the root canal access opening (Fig. 21.25A). The dentist places a rubber dam to isolate the discoloured tooth and removes all materials in the coronal portion of the tooth (i.e. access opening). The dentist removes guttapercha (to approximately 1–2 mm apical of the clinical crown) and enlarges the endodontic access opening sufficiently to ensure complete debridement of the pulp chamber (Fig. 21.25B).

Step 2: coronal sealing cement

Next, the dentist places an RMGI liner to seal the guttapercha of the root canal, filling from the coronal portion of the pulp chamber (Fig. 21.25C). After this seal has hardened, the dentist trims any excess material from the seal so that the discoloured dentin is exposed peripherally.

Step 3: sodium perborate bleaching

Sodium perborate is used with this technique because it is deemed extremely safe.[19] Using a cement spatula, with heavy pressure on a glass slab, one drop of saline or sterile anesthetic solution is blended with enough sodium perborate to form a creamy paste. A spoon excavator or similar instrument is used to fill the pulp chamber (with the bleaching mixture) to within 2 mm of the cavosurface margin, avoiding contact with the enamel cavosurface margins of the access opening.

Step 4: temporary seal

The dentist uses a cotton pellet to blot the mixture and places a temporary sealing material (e.g. Cavit [3M ESPE, St. Paul, MN] or Fuji IX [GC America, Alsip, IL]) to seal the access opening (Fig. 21.25D). The area should remain isolated for approximately 5 min after closure to evaluate the adequacy of the seal of the temporary restoration. If bubbles appear around the margins of the temporary material indicating leakage, the temporary restoration must be replaced. If no bubbles appear, the dentist removes the rubber dam and checks the occlusion to assess the presence or absence of contact on the temporary restoration.

Step 5: inter-appointment schedule

The sodium perborate should be changed *weekly*. On successful bleaching of the tooth, the chamber is rinsed and filled to within 2 mm of the cavosurface margin with a paste consisting of calcium hydroxide powder in sterile saline. (The enamel walls and margins are kept clean and free of the calcium hydroxide paste.) As noted earlier, a paste of calcium hydroxide powder and sterile water is placed immediately after bleaching in the pulp chamber to reduce the possibility of resorption.[20] The dentist reseals the access opening with a temporary restorative material, as previously described, and allows the calcium hydroxide material to remain in the pulp chamber for 2 weeks.

Step 6: final coronal restoration

Subsequently, the dentist removes the temporary restorative material, rinses away the calcium hydroxide and dries the pulp chamber. Next, the dentist restores the tooth with a light-cured composite (Fig. 21.25D).

Occasionally, a tooth that has been bleached by using the walking bleach technique and sealed with a composite restoration may subsequently become discoloured. In this instance, the alternative treatment option should be an attempt to bleach the tooth externally with one of the external bleaching techniques (see next section).

Clinical Considerations

- Although nonvital bleaching is effective, a slight potential (i.e. 1%) exists for a deleterious side effect termed *external cervical resorption*.[18] These sequela

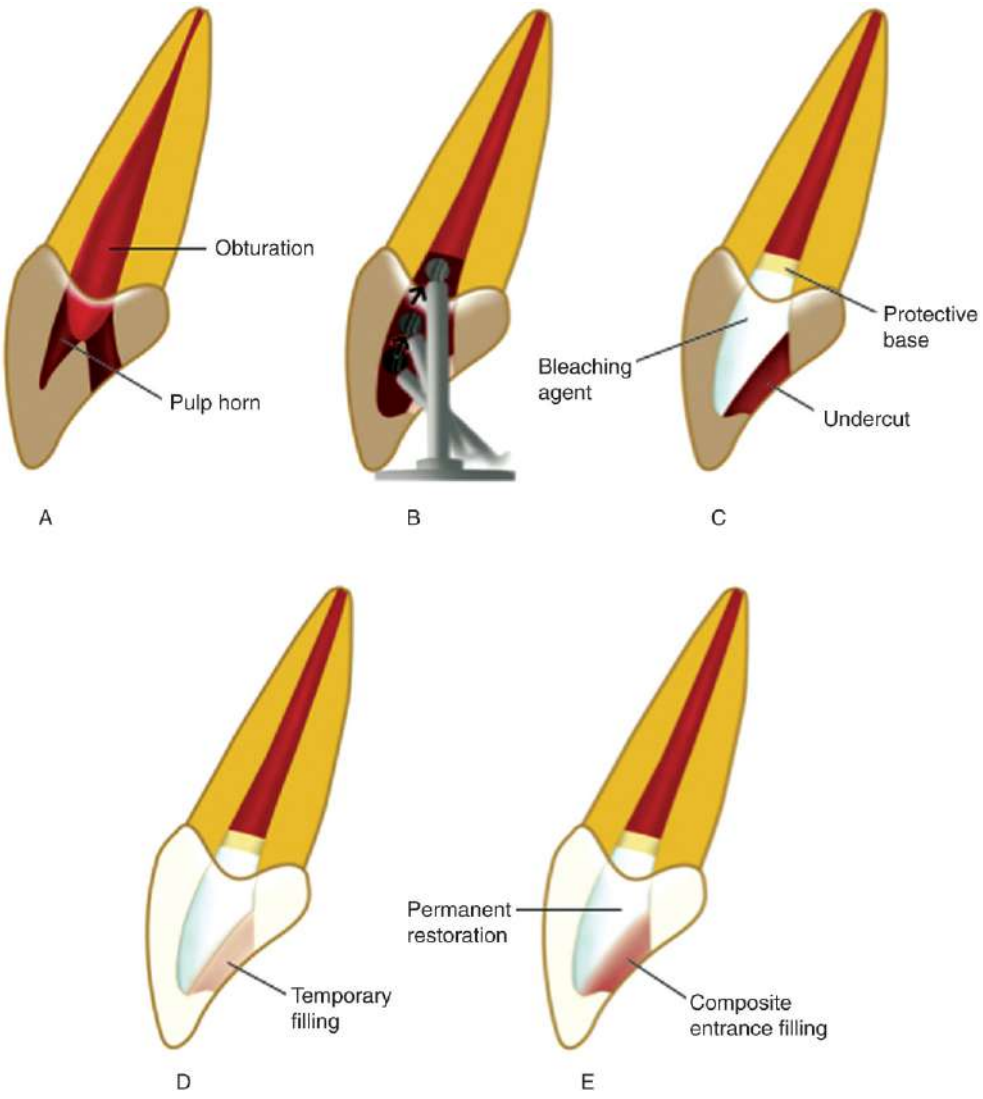


Figure 21.25 Walking bleach: (A) Nonvital discolouration of dentin. (B) The access cavity preparation is modified and gutta-percha is removed apically to just below the cervical margin. The pulp horns are cleaned with a round bur. (C) A protective *sealing cement* is placed over the gutta-percha, not extending above the cervical margin. A paste composed of sodium perborate and saline (mixed to the consistency of creamy paste) is placed. The incisal area is undercut to retain the temporary restoration. (D) A temporary filling seals the access. (E) A permanent restoration is placed when the desired bleaching effect has been achieved.
(From Chandra S, Gopikrishna V: *Grossman's endodontic practice*, ed 13, Philadelphia, PA, 2015, Wolters Kluwer-Lippincott.)

require prompt and aggressive treatment. In animal models, cervical resorption has been observed most when using a thermocatalytic technique with high heat.[21]

- The walking bleach technique or an in-office technique that does not require the use of heat is preferred for nonvital bleaching. To reduce the possibility of resorption, immediately after bleaching, a paste of calcium hydroxide powder and sterile water is placed in the pulp chamber as described in the following sections.[20]
- Also, sodium perborate alone, rather than in conjunction with hydrogen peroxide, should be used as the primary bleaching agent. Although sodium perborate may bleach more slowly, it is safer and less aggressive to the tooth.[22]
- Periodic radiographs should be obtained after bleaching to screen for cervical resorption, which generally has its onset in 1–7 years.[23]

II. Vital Bleaching Procedures

Generally the indications for the different vital bleaching techniques are similar, with patient preference, cost, compliance and difficulty in the removal of certain discolourations dictating the choice of treatment or combination of treatments.

Indications for Vital Bleaching

I. Teeth intrinsically discoloured because of ageing, trauma or certain medications.
II. External vital bleaching techniques are alternative treatment options for a failed, nonvital, walking bleach procedure.
III. Vital bleaching also is often indicated before and after restorative treatments to harmonize the shades of the restorative materials with those of natural teeth.
IV. Teeth exhibiting yellow or orange intrinsic discolouration seem to respond best to vital bleaching, whereas teeth exhibiting bluish grey discolourations often are considerably more difficult to treat in this manner.
V. Other indications for external bleaching include teeth that have been darkened by trauma but are still vital or teeth that have a poor endodontic prognosis because of the absence of a radiographically visible canal (i.e. calcific metamorphosis).
VI. Brown fluorosis stains also are often responsive to treatment, but white fluorosis stains may not be resolved effectively (although they can be made less obvious if the surrounding tooth structure can be significantly whitened).

Vital bleaching techniques include an

I. in-office vital bleaching technique referred to as *power bleaching* and
II. out-of-office vital bleaching alternative that is a 'dentist-prescribed, home-applied' technique (i.e. nightguard vital bleaching or simply 'at-home bleaching').[24,25]

These techniques may be used separately or in combination with one another. (Details are provided in subsequent sections.) Some over-the-counter bleaching materials, particularly products involving a trayless strip delivery system, also are effective for whitening teeth, but these are not discussed in this chapter.[26]

Overall, vital bleaching has been proven to be safe and effective when performed by or under the supervision of a dentist.

Clinical Considerations

- With short-term treatment, no appreciable effect has been observed on existing restorative materials, either in loss of material integrity or in colour change, with one exception: *Polymethyl methacrylate restorations exhibit a yellow-orange discolouration on exposure to carbamide peroxide*. For this reason, temporary crowns should be made from bis-acryl materials, rather than polymethyl methacrylate crown and bridge resin, if exposure to carbamide peroxide is anticipated.
- Because hydrogen peroxide has such a low molecular weight, it easily passes through enamel and dentin. This characteristic is thought to account for the *mild tooth sensitivity* occasionally experienced during treatment. This effect is transient, however, and no long-term harm to the pulp has been reported.

A. In-Office Vital Bleaching Technique

Clinical Technique

Step 1: rubber dam isolation

In-office vital bleaching requires an excellent rubber dam technique and careful patient management. Vaseline or cocoa butter may be placed on the patient's lips and gingival tissue before application of the rubber dam to help protect these soft tissues from any inadvertent exposure to the bleaching agent. Anterior teeth (and sometimes the first premolars) are isolated with a heavy rubber dam to provide maximum retraction of tissue and an optimal seal around teeth. A good seal of the dam is ensured by ligation of the dam with waxed dental tape or the use of a sealing putty or varnish. Light-cured, resin-based 'paint-on' rubber dam isolation media are available for use with in-office bleaching materials but cannot provide the same degree of protection and isolation as a conventionally applied rubber dam. Etching of teeth with 37% phosphoric acid, once considered a required part of this technique, is unnecessary.[27]

Step 2: 30–35% hydrogen peroxide bleaching

Numerous commercially available bleaching agents are available for in-office bleaching procedures. Most consist of paste or gel compositions that most commonly contain *30–35% hydrogen peroxide*. Other additives, such as metallic ion-producing materials or alkalinizing agents that can speed up the oxidation reaction, also are commonly found in these commercially available whitening products. The dentist places the hydrogen peroxide-containing paste or gel on teeth. The patient is instructed to report any sensations of burning of the lips or gingiva that would indicate a leaking dam and the need to terminate treatment.

Most of the credible research indicates that the *addition of light during the bleaching procedure* ***does not improve*** *the whitening result beyond what the bleach alone can achieve*.[28]

- Use of a light to generate heat may accelerate the oxidation reaction of the hydrogen peroxide and expedite treatment through a thermocatalytic effect. PAC lights and high-output quartz halogen lights have been commonly used for this purpose.
- Use of lights to heat the bleaching agent, however, causes a greater level of tooth dehydration. This effect not only can increase tooth sensitivity but also results in an immediate apparent whitening of the tooth owing to dehydration that makes the actual whitening result more difficult to assess.
- Use of carbon dioxide laser to heat the bleaching mixture and accelerate the bleaching treatment has not been recommended, according to a report of the American Dental Association because of the potential for hard or soft tissue damage.[29]

Step 3: duration of treatment

Contrary to the claims of some manufacturers, optimal whitening typically requires more than one bleaching treatment.[30] Bleaching treatments generally are rendered weekly for two to six treatments, with each treatment lasting 30–45 min.

Step 4: postoperative considerations

On completion of the treatment, the dentist rinses the patient's teeth, removes the rubber dam or isolation medium and cautions the patient about postoperative sensitivity. A nonsteroidal analgesic and anti-inflammatory drug may be administered if sensitivity is anticipated.

Patients may experience transient tooth sensitivity between appointments, but no long-term adverse effects of bleaching teeth with otherwise healthy pulps have been reported in the literature. Because the enamel is not acid etched, it is not necessary to polish the teeth after they have been bleached, and it is not essential to provide any fluoride treatment.

Advantages

I. This procedure is totally under the dentist's control and the soft tissue can be generally protected from the process.
II. This technique has the potential for bleaching teeth more rapidly.

Disadvantages

I. Cost of the treatment
II. Unknown duration of the treatment
III. The unpredictable outcome

B. Dentist-Prescribed, Home-Applied Technique

The dentist-prescribed, home-applied technique (i.e. nightguard vital bleaching) is much less labour intensive and requires substantially less in-office time.

Clinical Technique

Step 1: fabrication of vinyl custom nightguard

An impression of the arch to be treated is made and poured in cast stone. The nightguard is formed on the cast with the use of a heated vacuum-forming machine. After the machine has warmed up for 10 min, a sheet of 0.75-to-1.5-mm (0.020-to-0.040-in) soft vinyl nightguard material is inserted and allowed to be softened by heat until the material sags approximately by 25 mm (1 in). The top portion of the machine is closed slowly and gently, and the vacuum is allowed to form the heat-softened material around the cast. After sufficient time has been allowed for adaptation of the material, the operator turns off the machine and allows the material to cool.

Next, the operator uses scissors or a No. 11 surgical blade in a Bard-Parker handle to trim in a smooth, straight cut about 3–5 mm from the most apical portion of the gingival crest of teeth (facially and lingually). This excess material is removed first. The horseshoe-shaped nightguard is removed from the cast. The operator trims the facial edges of the nightguard in a scalloped design, following the outline of the free gingival crest and using sharp, curved scissors. Scalloping of the lingual surface is optional because the bleaching material is applied primarily to the facial aspects of teeth. Alternatively (on the lingual aspect), the nightguard may be trimmed apically to within 2 mm of the free gingival crest in a smooth, horseshoe-shaped configuration. This scalloped design is preferred because it allows the tray to cover only teeth and prevents entrapment of the bleaching material between gingival tissue and the nightguard. The nightguard is completed and is ready for delivery to the patient (Fig. 21.26).

Step 2: evaluation of the custom nightguard

The dentist inserts the nightguard into the patient's mouth and evaluates it for adaptation, rough edges or blanching of tissue. A properly fitting nightguard is shown in Fig. 21.27 Further shortening (i.e. trimming) may be indicated in problem areas. The dentist evaluates the occlusion on the nightguard with the patient in maximum intercuspation. If the patient is unable to obtain a comfortable occlusion because of premature posterior tooth contacts, the nightguard is trimmed to exclude coverage of the terminal posterior teeth, as needed (to allow optimal tooth contact in maximum inter-cuspation). In addition, if no lingual scalloping is done, the edges of the guard on the palate should terminate in grooves or valleys, where possible, rather than on the heights of soft tissue contours (e.g. in the area of the incisive papilla).

Step 3: carbamide peroxide bleaching

A *10%–15% carbamide peroxide bleaching material* generally is recommended for this bleaching technique. Commercial bleaching products are available as clear gels and white pastes.

Carbamide peroxide degrades into *3% hydrogen peroxide (active ingredient)* and *7% urea*. Bleaching materials containing *carbopol* are recommended because carbopol thickens the bleaching solution and extends the oxidation process. On the basis of numerous research studies, carbamide peroxide bleaching materials seem to be safe and effective when administered by or under the supervision of a dentist.[22]

Step 4: clinical use of nightguard bleaching

- The patient is instructed in the application of the bleaching gel or paste into the nightguard. A thin bead of material is extruded into the nightguard



Figure 21.26 (A) Upper and lower casts trimmed to maintain a base height of 10–12 mm. (B) Reservoirs are formed on the labial surfaces of the teeth. (C) The margins of the reservoir should end 1 mm short of the free gingival margin. (D) The cast is duplicated using irreversible hydrocolloid impression material. (E) The cast is placed on the vacuum-forming machine. Note the plastic tray material is warmed from a coil above. The plastic tray material sags as it is softened. Vacuum is created at the base of the machine and adapts the tray material on the cast. (F) The free gingival margin is marked and trimmed accordingly. (G) Bleaching tray has to be tried in the patient's mouth and necessary adjustment should be made before delivering the tray.
(From Chandra S, Gopikrishna V: *Grossman's endodontic practice*, ed 13, Philadelphia, PA, 2015, Wolters Kluwer-Lippincott.)

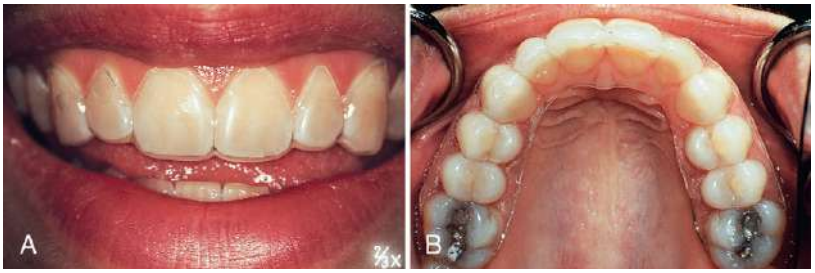


Figure 21.27 Nightguard for vital bleaching. (A and B) Clear plastic nightguard properly seated and positioned in the mouth (scalloped on facial, unscalloped on lingual).

along the facial aspects corresponding to the area of each tooth to be bleached. Usually, only the anterior six to eight teeth are bleached. The clinician should review proper insertion of the nightguard with the patient. After inserting the nightguard, excess material is wiped from the soft tissue along the edge with a soft-bristled toothbrush. No excess material should be allowed to remain on soft tissue because of the potential for gingival irritation. The patient should be informed not to drink liquids or rinse during treatment and to remove the nightguard for meals and oral hygiene.

- Although no single treatment regimen is best for all patients, most patients prefer an overnight treatment approach because of the convenience. If the nightguard is worn at night, a single application of bleaching material at bedtime is indicated. In the morning, the patient should remove the nightguard, clean it under running water with a toothbrush, and store it in the container provided. *Total treatment time using an overnight approach is usually 1–2 weeks*. If patients cannot tolerate overnight bleaching, the bleaching time and frequency can be adjusted to accommodate the patient's comfort level. In addition, in these cases, tolerance to the nightguard and bleaching material generally are improved if the patient gradually increases the wearing time each day.
- If either of the two primary adverse effects occurs (i.e. *sensitive teeth* or *irritated gingiva*), the patient should reduce or discontinue treatment immediately and contact the dentist so that the cause of the problem can be determined and the treatment approach modified. The dentist may prescribe desensitizing agents to help alleviate sensitivity associated with bleaching.
- It is recommended that *only one arch be bleached at a time*, beginning with the maxillary arch. Bleaching the maxillary arch first allows the untreated mandibular arch to serve as a constant standard for comparison. Restricting the bleaching to one arch at a time reduces the potential for occlusal problems that could occur if the thicknesses of two mouthguards were interposed simultaneously. Fig. 21.28 illustrates a typical case before and after treatment with nightguard vital bleaching.

Advantages

I. The use of a lower concentration of peroxide (generally *10%–15% carbamide peroxide*)
II. Ease of application
III. Minimal side effects
IV. Lower cost because of the reduced chair time required for treatment

Disadvantages

I. Reliance on patient compliance
II. Longer treatment time
III. The (unknown) potential for soft tissue changes with excessively extended use

Bleaching of Tetracycline-Stained Teeth

Tetracycline-stained teeth typically are much more *resistant to bleaching*. Teeth stained with tetracycline require prolonged treatment durations of several months before any results are observed. Often, tetracycline-stained teeth are

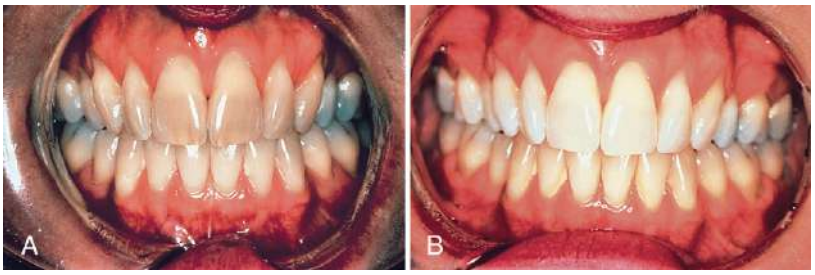


Figure 21.28 Nightguard vital bleaching. (A) Before bleaching treatment and (B) after treatment.

unresponsive to the procedure, especially if the stains are blue-grey in colour. Tetracycline-stained teeth may approach but never seem to achieve the appearance of normal teeth. A single tetracycline-stained tooth with previous endodontic therapy or a different pulp size may respond differently from other teeth in the arch to the bleaching technique.

- Because bleaching tetracycline-stained teeth is difficult, some clinicians advocate intentional endodontic therapy along with the use of an intra-coronal nonvital bleaching technique to overcome this problem (Fig. 21.29). Although the esthetic result appears much better than that obtained from external bleaching, this approach involves all the inherent risks otherwise associated with endodontic treatment.
- External bleaching techniques offer a safer alternative, although they may not be as rapid or effective.
- Veneers or full crowns are alternative esthetic treatment methods for difficult tetracycline-stained teeth but involve irreversible restorative techniques (see section Indirect Veneer Techniques).
- No one bleaching technique is effective in all cases, and all successes are not equal. Often, with vital bleaching, a combination of the in-office technique and the dentist-prescribed, home-applied technique has better results than either technique used alone.

Micro-Abrasion and Macro-Abrasion

Micro-abrasion and *macro-abrasion* represent conservative alternatives for the reduction or elimination of superficial discolourations. As the terms imply, the stained areas or defects are abraded away. *These techniques result in the physical removal of the tooth structure and are indicated only for stains or enamel defects that do not extend beyond a few tenths of a millimetre in depth*. If the defect or discolouration remains even after treatment with micro-abrasion or macro-abrasion, a restorative alternative is indicated.

Micro-Abrasion

- In 1984, McCloskey reported the use of 18% hydrochloric acid swabbed on teeth for the removal of superficial fluorosis stains.[31]
- Subsequently, in 1986, Croll and Cavanaugh modified the technique to include the use of pumice with 18% hydrochloric acid to form a paste applied with a tongue blade.[32] This technique is called *micro-abrasion* and involves the surface dissolution of the enamel by acid along with the abrasiveness of the pumice to remove superficial stains or defects.
- Since that time, Croll further modified the technique:
  - reducing the concentration of the hydrochloric acid to approximately 11% and
  - Increasing the abrasiveness of the paste using silicon carbide particles (in a water-soluble gel paste) instead of pumice.[33]

This product, marketed as Prema compound (Premier Dental Products Co., Plymouth Meeting, PA) or Opalustre (Ultradent Products, Inc., South Jordan, UT), represents an improved and safer means for the removal of superficial stains or defects. This technique involves the physical removal of tooth structure and does not remove stains or defects through any bleaching phenomena.

Clinical Considerations

- Before treatment, the clinician should evaluate the nature and extent of the enamel defect or stain
- The dentist should be able to differentiate between nonhereditary developmental dysmineralization (i.e. abnormal mineralization) defects (e.g. white or light brown fluoretic enamel and the idiopathic white or light brown spot) versus incipient carious lesions.
- Incipient carious lesions usually are located near the gingival margin. *Incipient* carious *lesions have a smooth surface and appear opaque or chalky white when dried but are less visible when hydrated.* Incipient caries is reversible if treated immediately. Changing the oral environment through oral hygiene practices and dietary adjustments allows remineralization to occur.
- If the caries lesion has progressed to have a slightly roughened surface, however, micro-abrasion coupled with a remineralization program is an initial option. If this approach is unsuccessful, it can be followed by a restoration. Cavitation of the enamel surface is an indication for restorative intervention. As the location of smooth-surface enamel caries nears the cementoenamel



Figure 21.29 Bleaching tetracycline-stained teeth. (A) Before nonvital bleaching and (B) after treatment. (Courtesy: Dr. Wayne Mohorn.)

junction, then enamel is too thin to permit micro-abrasion or macro-abrasion as a treatment option.

- A developmental discoloured spot (opaque white or light brown) is the result of an unknown, local traumatic event during amelogenesis and is termed *idiopathic*. Its surface is intact, smooth and hard. It usually is located in the incisal (occlusal) half of enamel, which contributes to the unsightly appearance. The patient (or the patient's parents in the case of a child) must be informed that an accurate prognosis for micro-abrasion cannot be given but that micro-abrasion will be applied first. If micro-abrasion is unsuccessful because of the depth of the defect exceeding 0.2–0.3 mm, the tooth will be restored with a tooth-coloured restoration.
- Surface discolourations resulting from fluorosis also can be removed by micro-abrasion if the discolouration is within the 0.2–0.3-mm removal depth limit.

Clinical Procedure

Fig. 21.30A shows a young patient with fluorosis stains on teeth Nos. 8 and 9. A rubber dam is placed to isolate the teeth to be treated and to protect the gingival tissues from the acid in the Prema paste or compound (Premier Dental Products). Protective glasses should be worn by the patient to shield the eyes from any spatter. The Prema paste is applied to the defective area of the tooth with a special rubber cup that has fluted edges (Fig. 21.30B and C). The abrasive compound can be applied with either the side or the end of the rubber cup. A 10× gear reduction, low-speed handpiece (similar to that used for placing pins) is recommended for the application of the Prema compound to reduce the possibility of removing too much tooth structure and to prevent spatter. Moderately firm pressure is used in applying the compound.

For small, localized, idiopathic white or light brown areas, a hand application device also is available for use with the Prema compound (Fig. 21.30D). Periodically, the paste is rinsed away to assess the extent of defect removal. The facial surface also is viewed with a mirror from the incisal aspect to determine how much tooth structure has been removed. Care must be taken not to remove too much tooth structure. The procedure is continued until the defect is removed or until it is deemed imprudent to continue further (Fig. 21.30E). The treated areas are polished with a fluoride-containing prophy paste to restore surface lustre (Fig. 21.30F). Immediately after treatment, a topical fluoride is applied to teeth to enhance remineralization (Fig. 21.30G). Results are shown in Fig. 21.30H.

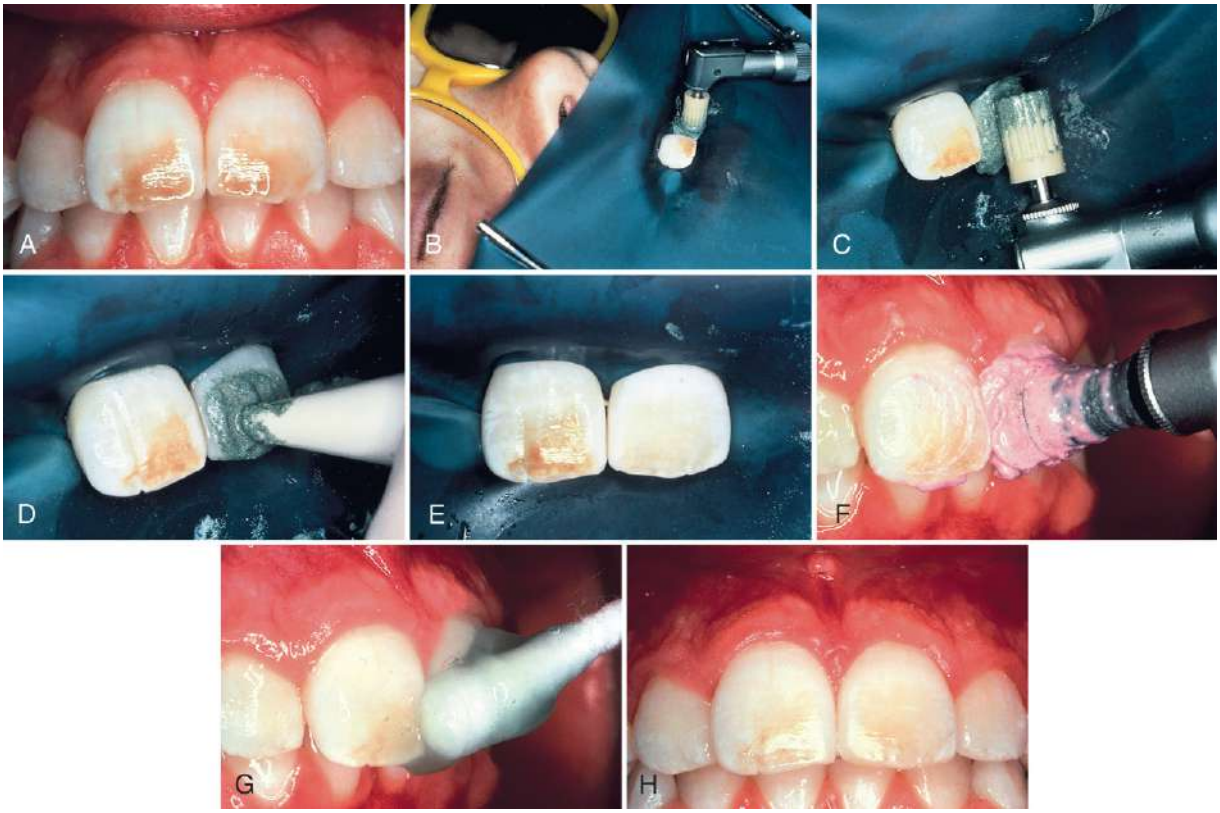


Figure 21.30 Micro-abrasion. (A) Young patient with unesthetic fluorosis stains on central incisors. (B and C) Prema compound applied with special rubber cup with fluted edges. Protective glasses and rubber dam are needed for the safety of the patient. (D) Hand applicator for applying Prema compound. (E) Stain removed from the left central incisor after micro-abrasion. (F) Treated enamel surfaces polished with prophylactic paste. (G) Topical fluoride applied to treated enamel surfaces. (H) Final esthetic result.
(Courtesy: Dr. Ted Croll.)

Macro-Abrasion

An alternative technique for the removal of localized, superficial white spots (not subject to conservative, remineralization therapy) and other surface stains or defects is called *macro-abrasion*.

Clinical Procedure

- Macro-abrasion simply uses a *12-fluted composite finishing bur or a fine-grit finishing diamond* in a high-speed handpiece to remove the defect (Fig. 21.31A and B).
- Care must be taken to use light, intermittent pressure and to monitor the removal of tooth structure carefully to avoid irreversible damage to the tooth. Air-water spray is recommended, not only as a coolant but also to maintain the tooth in a hydrated state to facilitate the assessment of defect removal. Teeth that have white spot defects are particularly susceptible to dehydration resulting in other apparent white spots that are not normally seen when the tooth is hydrated. Dehydration exaggerates the appearance of white spots and makes defect removal difficult to assess.
- After removal of the defect or on termination of any further removal of tooth structure, a *30-fluted finishing bur* is used to remove any facets or striations created by the previous instruments.
- Final polishing is accomplished with an abrasive rubber point (Fig. 21.31C). The results are shown in Fig. 21.31D.

Micro-Abrasion versus Macro-Abrasion

Comparable results can be achieved with micro-abrasion and macro-abrasion. Both treatments have advantages as well as disadvantages.

- Micro-abrasion has the advantage of ensuring better control of the removal of tooth structure.
- High-speed instrumentation used in macro-abrasion is technique sensitive and can have catastrophic results if the clinician fails to use extreme caution.
- Macro-abrasion is considerably faster and does not require the use of a rubber dam or special instrumentation. Defect removal also is easier with macro-abrasion compared with micro-abrasion if an air-water spray is used during treatment to maintain hydration of teeth.

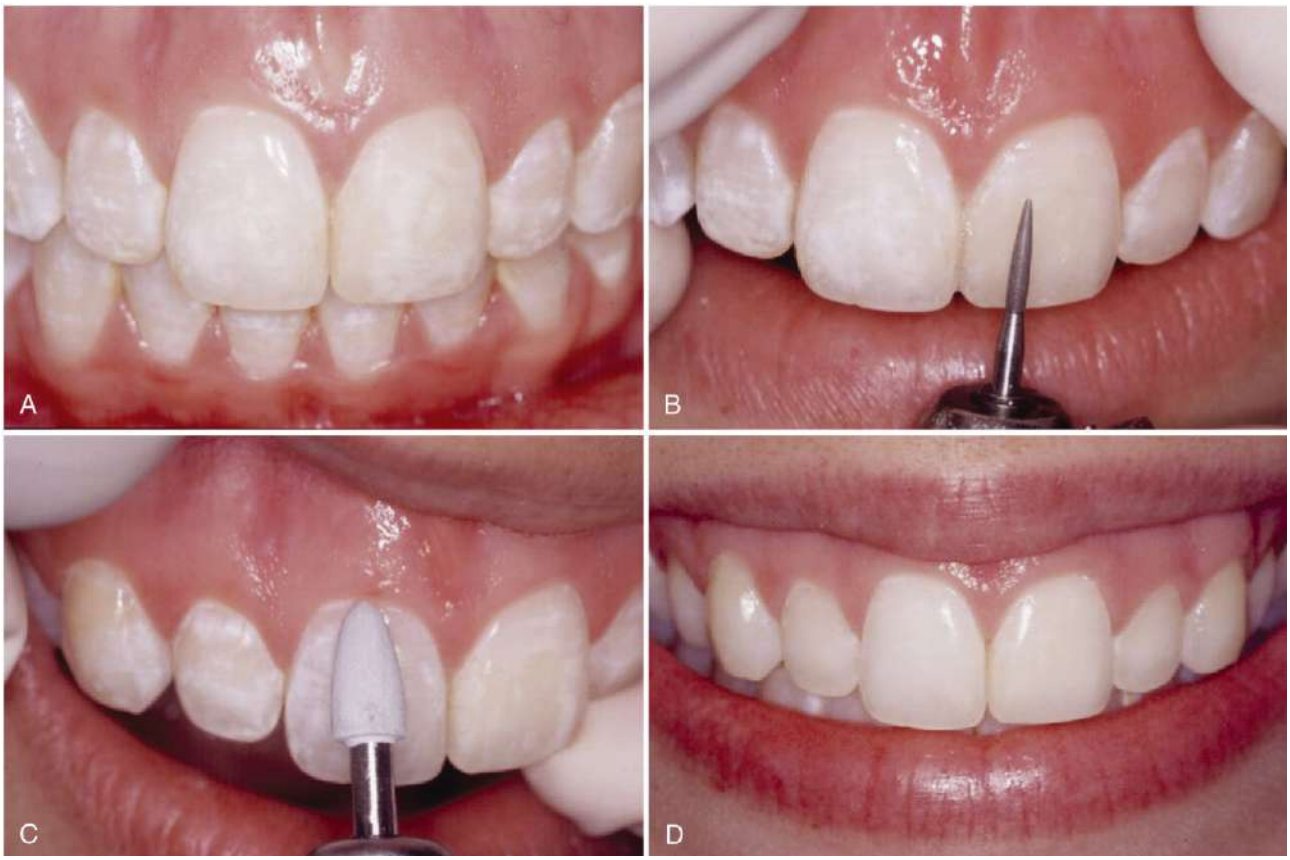


Figure 21.31 Macro-abrasion. (A) Outer surfaces of maxillary anterior teeth are unesthetic because of superficial enamel defects; (B and C) removal of discolouration by abrasive surfacing and polishing procedures and (D) completed treatment revealing conservative esthetic outcome.

- Nonetheless, micro-abrasion is recommended over macro-abrasion for the treatment of superficial defects in children because of better operator control and superior patient acceptance.
- To accelerate the process, a combination of macro-abrasion and micro-abrasion also may be considered. Gross removal of the defect is accomplished with macro-abrasion, followed by final treatment with micro-abrasion.

Veneers

A veneer is a layer of tooth-coloured material that is applied to a tooth to restore localized or generalized defects and intrinsic discolourations (e.g. Figs. 21.7,21.34,21.35,21.36 and 21.42).

Typically, veneers are made of directly applied composite, processed composite, porcelain or pressed ceramic materials.

Indications

Common indications for veneers (Fig. 21.32) include facial surfaces of

- malformed teeth,
- discoloured teeth,
- abraded or eroded teeth and
- teeth with faulty restorations.

The various types of veneers is given in Box 21.2.

The main two types of esthetic veneers (Fig. 21.33) are

I. **Partial veneers**. Partial veneers are indicated for the restoration of localized defects or areas of intrinsic discolouration (Fig. 21.34).

II. **Full veneers**. Full veneers are indicated for the restoration of generalized defects or areas of intrinsic staining involving most of the facial surface of the tooth (e.g. Figs. 21.7,21.36 and 21.37).

Box 21.2 Types of Veneers

I. Based on the *extent of the tooth involved*, veneers can be classified as
   i. Partial veneers
   ii. Full veneers:
      - Window preparation
      - Butt joint incisal preparation
      - Incisal lapping preparation

II. Based on the *type of material employed,* veneers can be classified as
   i. Directly applied composite veneer
   ii. Processed composite veneer
   iii. Feldspathic porcelain veneer
   iv. Pressed ceramic veneer

III. Based on the *mode of fabrication* veneers can be classified into
   i. Direct veneers
      - Direct partial veneers
      - Direct full veneers
   ii. Indirect veneers
      - No-prep veneers
      - Etched porcelain veneers
      - Pressed ceramic veneers



Figure 21.32 Clinical examples of indications for treatment with veneers include teeth affected by tetracycline drug staining (A); fluorosis or enamel hypoplasia (B); acid-induced erosion (e.g. lemon-sucking habit) (C) and defective or improperly done existing veneers (note significant gingival overhang with associated purulent exudate) (D).

Clinical Considerations

The preparation and restoration of a tooth with a veneer should be carried out in a manner that provides optimal function, esthetics, retention, physiologic contours and longevity. All of these objectives should be accomplished without compromising the strength of the remaining tooth structure. If the veneer becomes chipped, discoloured or worn, it usually can be repaired or replaced.

Patient Factors

Several important factors, including patient age, occlusion, tissue health, position and alignment of teeth and oral hygiene, must be evaluated before pursuing full veneers as a treatment option.

If full veneers are done, care must be taken to provide proper physiologic contours, particularly in the gingival area, to ensure good gingival health. An example of poorly contoured veneers is shown in Fig. 21.32D; severe gingival irritation exists around the over-contoured veneers; a purulent exudate is evident on probing the margins with an explorer.

Indirect versus Direct Full Veneers

Full veneers can be accomplished by the direct or indirect technique.

- *Direct full veneer*:
  - When only a few teeth are involved or when the entire facial surface is not faulty (i.e. partial veneers), directly applied composite veneers can be completed chairside in one appointment for the patient.
  - Placing direct composite full veneers is time consuming and labour intensive.
  - For cases involving young children or a single discoloured tooth, or when the patient's time or money is limited, precluding a laboratory-fabricated veneer, the direct technique is a viable option.
- *Indirect full veneer*:
  Indirect veneers require two appointments but typically offer three advantages over directly placed full veneers:
  - Indirectly fabricated veneers are much less sensitive to operator technique. Considerable artistic expertise and attention to detail are required to consistently achieve esthetically pleasing and physiologically sound direct veneers. Indirect veneers are made by a laboratory technician and are typically more esthetic.
  - If multiple teeth are to be veneered, indirect veneers usually can be placed much more expeditiously.
  - Indirect veneers typically last much longer than do direct veneers, especially if they are made of porcelain or pressed ceramic.

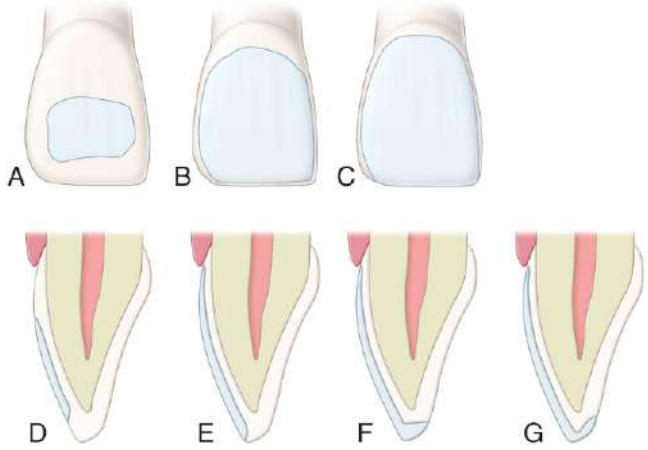


Figure 21.33 Four types of veneers. (A) Facial view of partial veneer that does not extend sub-gingivally or involve the incisal angle. (B) Full veneer with window preparation design that extends to the gingival crest and terminates at the facioincisal angle. (C) Full veneer with either a butt-joint incisal preparation design or an incisal-lapping preparation design extending sub-gingivally and including all of incisal surface. (Sub-gingival extension is indicated only for preparation of darkly stained teeth and is not considered routine.) (D–G) Cross-sections of the four types of veneers: partial veneer (D), full veneer with window preparation design (E), full veneer with butt-joint incisal preparation design (F) and full veneer with incisal-lapping incisal preparation design (G).

Extent of Preparation Depth

Some controversy exists regarding the extent of tooth preparation that is necessary and the amount of coverage for both direct and indirect fabricated veneers (Fig. 21.33).

Acid etching without any tooth preparation (no prep)

Some operators prefer to etch the existing enamel and apply the veneer (direct or indirect type) to the entire existing facial surface without any tooth preparation. The perceived advantage of these 'no prep' veneers is that little or no tooth structure is removed. Also, in the event of failure or if the patient does not like the veneer, it supposedly can be removed (being reversible), although in actuality, it is never easy to remove any bonded veneer without concomitant removal of some tooth structure (see also section No-Prep Veneers).

Several significant problems exist, however, with this approach:

- First, to achieve an esthetic result, if the tooth is otherwise of normal contour, the facial surface of such a restoration will be over-contoured, appearing and feeling unnatural. This observation is true for both direct and indirect veneers. An over-contoured veneer frequently results in gingival irritation with accompanying hyperemia and bleeding caused by bulbous and impinging gingival contours.

- Second, with regard to direct veneers, the veneer is more likely to be dislodged when no tooth structure is removed before the etching and bonding procedures are done. If the veneer is lost, it can be replaced. The patient may live in constant fear, however, that it will happen again, possibly causing embarrassment.

The reversibility of no-prep veneers may seem desirable and appealing to patients from a psychological standpoint; however, few patients who elect to have veneers wish to return to the original condition. In addition, removing full veneers with no damage to the underlying unprepared tooth, as noted earlier, is exceedingly difficult if not impossible.

Intra-enamel preparation

To achieve esthetically pleasing and physiologically sound results consistently, an intra-enamel preparation is usually indicated. The only exception is in cases in which the facial aspect of the tooth is significantly under-contoured because of severe abrasion or erosion. In these cases, mere roughening of the involved enamel and defining of the peripheral margins are indicated.

Intra-enamel preparation (or the roughening of the surface in under-contoured areas) before placing a veneer is strongly recommended for the following reasons:

I. To provide space for bonding and veneering materials for maximal esthetics without over-contouring
II. To remove the outer, fluoride-rich layer of enamel that may be more resistant to acid etching
III. To create a rough surface for improved bonding
IV. To establish a definite finish line

Establishing an intra-enamel preparation with a definite finish line is particularly important when placing indirectly fabricated veneers. Accurate positioning and seating of an indirectly made veneer are enhanced significantly if an intra-enamel preparation is present.

Location of Gingival Margin of Veneer

Another controversy involves the location of the gingival margin of the veneer (Fig. 21.33). Should it terminate short of the free gingival crest at the level of the gingival crest or apical of the gingival crest? The answer depends on the individual situation.

I. If the *defect or discolouration does not extend sub-gingivally*, the margin of the veneer should not extend sub-gingivally: The position of the gingival margin of any veneer is dictated by the extent of the defects or discolouration and the amount of tooth structure that is visible with maximum smiling. If a patient exhibits a high smile line that exposes the entire facial surface of the tooth and if defects like fluorosis stains are generalized, then the margin of the veneer must be positioned at the level of the crest of tissue to optimize esthetics.
II. However, the only logical reason for extending the margin sub-gingivally is if the *gingival area is carious or defective*, warranting restoration, or if it involves significantly dark discolouration that presents a difficult esthetic problem. No restorative material is as good as normal tooth structure, and the gingival tissue is never as healthy when it is in contact with an artificial material.

Role of Full Veneer Preparation Designs

Three basic preparation designs exist for full veneers:

I. *Window preparation*:
   - A window preparation is recommended for most direct composite veneers. It is also frequently used in cases of indirectly fabricated veneers where the outline form of the canine is intact and the patient is canine guided. This intra-enamel preparation design preserves the functional lingual and incisal surfaces of maxillary anterior teeth, protecting the veneers from significant occlusal stress. This quality is of particular importance with direct composite veneers.
   - A window preparation design also is recommended for indirectly fabricated porcelain veneers in patients who exhibit a canine-guided pattern of lateral guidance and in whom the maxillary canines are of normal contour with little incisal wear or notching. By using a window preparation, the functional surfaces are better preserved in enamel (Fig. 21.33E). This design reduces the potential for accelerated wear of the opposing tooth that could result if the functional path involved porcelain on the lingual and incisal surfaces, as with incisal butt-joint or incisal-lapping designs.
II. *Butt-joint incisal preparation*:
   - For most indirectly fabricated porcelain veneers, either a butt-joint incisal design or an incisal-lapping approach is used.
   - A butt-joint incisal design is used routinely in cases where no defects exist along the lingual aspect of the incisal edge.
   - It is the simplest design and is used to easily provide adequate reduction of the tooth to accommodate the needed strength of the porcelain veneer in this area of the preparation (Fig. 21.33F).
III. *Incisal-lapping preparation*:
   - An incisal-lapping preparation (Fig. 21.33) is indicated when :
     – the tooth being veneered needs lengthening or
     – when an incisal defect warrants restoration.
   - The extent of the lapping onto the lingual surface is generally dictated by the extent of the lingual incisal defect or by the amount of faciolingual resistance form desired for reinforcement of the incisal edge (Fig. 21.33G).

Clinical Limitations

- Darkly stained teeth, especially *teeth discoloured by tetracycline*, are much more difficult to veneer with full veneers compared with teeth with generalized defects but that have normal underlying colouration (Fig. 21.45). The difficulty is compounded further when the cervical areas are badly discoloured. Usually, only the six maxillary anterior teeth require correction because they are the most noticeable when a person smiles or talks. The maxillary first premolars (and to a lesser extent the second premolars) also are included, however, if they too are noticeable on smiling.
- *Discoloured mandibular anterior teeth* are rarely indicated for veneers because the facioincisal portions are thin and usually subject to biting forces and attrition. Veneering mandibular teeth is discouraged if the teeth are in normal occlusal contact because it is exceedingly difficult to achieve adequate reduction of the enamel to compensate for the thickness of the veneering material. Also, if porcelain veneers are placed on mandibular teeth, the veneers may accelerate the wear of opposing maxillary teeth because of the abrasive nature of the porcelain. In most cases, the lower lip hides these teeth so esthetics is not a significant problem. Most patients are satisfied with the conservative approach of veneering only maxillary anterior teeth.

Direct Veneer Technique

I. Direct Partial Veneers

Small localized intrinsic discolourations or defects that are surrounded by healthy enamel are ideally treated with direct partial veneers (Fig. 21.1B and C). Often practitioners place full veneers when only partial veneers are indicated. The four anterior teeth in Fig. 21.34A illustrate the clinical technique for placing partial veneers. These defects can be restored in one appointment with a light-cured composite.

Fig. 21.34A shows four anterior teeth that had previously received direct composite veneers with no enamel preparation for the restoration of developmental white spot lesions. However, even after veneering, the white spots still show through the veneers. On removal of the defective veneers, the localized white spots are evident (Fig. 21.34B). Models illustrating proper preparation are shown in Fig. 21.34C.

Clinical Technique

- Preliminary steps include cleaning, shade selection and isolation with cotton rolls or rubber dam.
- Anaesthesia usually is not required unless the defect is deep, extending into dentin.
- The outline form is dictated solely by the extent of the defect and should include all discoloured areas.
- The clinician should use a coarse, elliptical or round diamond instrument with air-water coolant to remove the defect. The use of water-air spray is also imperative so that the tooth can be maintained in a hydrated state. If dehydration is allowed to occur, it can cause the appearance of other white spots which are artefacts and which will make defect assessment much more difficult (Fig. 21.34D).
- After preparation, etching and restoration of the defective areas (as described in the following paragraph), the finished partial veneers are seen (Fig. 21.34E).

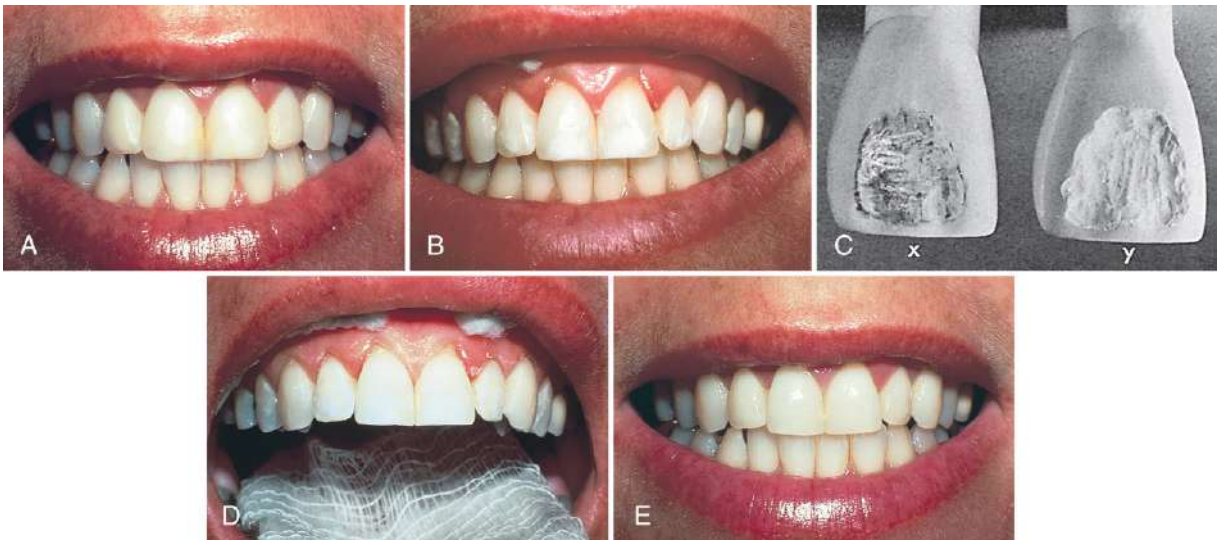


Figure 21.34 Direct partial veneers. (A) Patient with over-contoured direct full veneers. (B) After removal of old veneer, localized white spots are evident. (C) Models illustrate fault (*x*) and cavity preparation (*y*). The chamfered margins are irregular in outline. (D) Intra-enamel preparations for partial veneer restorations. (E) Conservative esthetic result of completed partial veneers.

- Usually, it is desirable to remove all of the discoloured enamel in a pulpal direction. If the entire defect or stain is removed, a *microfilled composite* is recommended for restoring the preparation. Microfills are excellent 'enamel replacement' materials because of their optical properties. If the tooth has been maintained in a hydrated state, the microfilled composite can be positioned on a trial basis to assess the accuracy of the shade prior to final restoration.
- *Nanofilled composites* also are excellent material choices for this technique. If a residual lightly stained area or white spot remains in enamel, however, an intrinsically less translucent composite can be used rather than extending the preparation into dentin to eliminate the defect. Most composites filled primarily with radiopaque fillers (e.g. barium glass) are more optically opaque with intrinsic masking qualities (in addition to being radiopaque). Use of these types of composites for the restoration of preparations with light, residual stains is most effective and conserves the tooth structure. In this example, all restorations are of a light-cured microfilled composite.

II. Direct Full Veneers

Placing direct full composite veneers is very time consuming. Extensive enamel hypoplasia involving all maxillary anterior teeth was treated by placing direct full veneers in this case (Fig. 21.35A). A diastema also was present between the central incisors. The patient desired to have the hypoplasia and the diastema corrected; examination indicated a good prognosis. A direct technique was used with a light-cured microfilled composite. Although all six teeth can be restored at the same appointment, it may be less traumatic for the patient and the dentist if the veneers are placed in two appointments. In this example, the central incisors were completed during the first appointment, and the lateral incisors and canines were completed during the second appointment.

Clinical Technique

- After the teeth to receive the veneers are cleaned and a shade is selected, the area is isolated with cotton rolls and retraction cords. Both central incisors are prepared with a coarse, rounded-end diamond instrument.
- The *window preparation* typically is made to a depth roughly equivalent to half the thickness of the facial enamel, ranging from approximately 0.5–0.75 mm midfacially and tapering down to a depth of about 0.3–0.5 mm along the gingival margin, depending on the thickness of enamel (Fig. 21.33).
- A *well-defined chamfer* at the level of the gingival crest provides a definite preparation margin for subsequent finishing procedures.
- The margins are not extended sub-gingivally because these areas are not defective.
- The preparation for all veneer types (both direct and indirect) normally is terminated just *facial to the proximal contact* except in the case of a diastema (Fig. 21.35B).
- When *interdental spaces* exist, the preparations must be extended from the facial surfaces onto the mesial surfaces, terminating at the mesiolingual line angles (Fig. 21.35C and D). This lingual extension of the

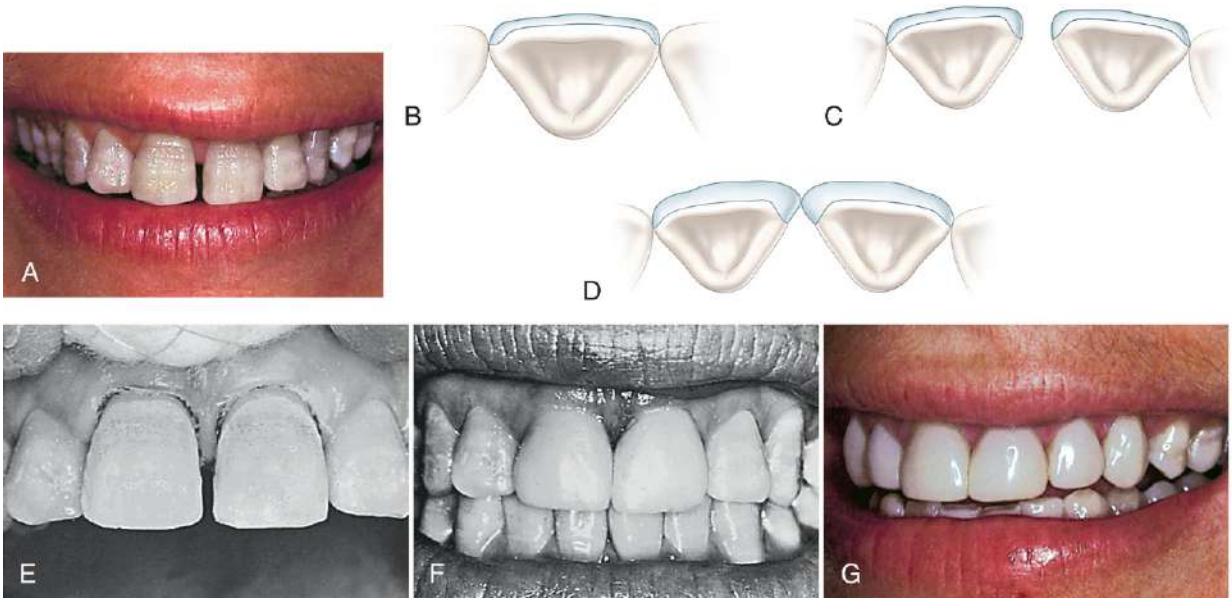


Figure 21.35 Direct full veneers using light-cured composite. (A) Enamel hypoplasia of maxillary anterior teeth. (B) Typical preparation of facial surface for full veneer. (C) The preparation is extended onto the mesial surface to allow closure of diastema. (D) Full veneers restore proximal contact. (E) Etched preparations of central incisors. (F) Veneers completed on maxillary central incisors. (G) Treatment completed with placement of full veneers on remaining maxillary anterior teeth.

preparation allows for proper reestablishment of the entire proximal contour of the tooth in the final restoration.

- The incisal edges were not included in the preparations in this example because no discolouration was present. In addition, preservation of the incisal edges better protects the veneers from heavy functional forces, as noted earlier for window preparations.
- The teeth to be treated should be restored one at a time. After the etching, rinsing and drying procedures (Fig. 21.35E), the dentist applies and polymerizes the resin-bonding agent. The dentist places the composite on the tooth in increments, especially along the gingival margin, to reduce the effects of polymerization shrinkage.
- The composite is placed in slight excess to allow some freedom in contouring. It is helpful to inspect the facial surface from an incisal view with a mirror to evaluate the contour before polymerization. After the first veneer is finished, the second tooth is restored in a similar manner (Fig. 21.35F). In this case, the remaining four anterior teeth are restored with direct composite veneers (Fig. 21.35G) at the second appointment. Chapter 19 describes the procedures used to insert and finish composite restorations. Another excellent example of direct composite veneers is seen in Fig. 21.36.

Indirect Veneer Techniques

Many dentists find that the preparation, placement and finishing of several direct veneers at one time is too difficult, fatiguing and time consuming. Some patients become uncomfortable and restless during long appointments. In addition, veneer shades and contours can be better controlled when made outside of the mouth on a cast. For these reasons, indirect veneer techniques are usually preferable.

Indirect veneers are primarily made of

I. processed composite,
II. feldspathic porcelain, and
III. cast or pressed ceramic.

Because of superior strength, durability and conservation of the tooth structure, feldspathic porcelain bonded to intra-enamel preparations has historically been the preferred approach for indirect veneering techniques used by dentists. Some pressed ceramic veneering materials offer comparable esthetic qualities but may require a



Figure 21.36 Direct full veneers using light-cured composite for defective veneers. (A) Defective composite veneers with marginal staining; (B) conservative intra-enamel preparation; and (C) new direct composite veneers on maxillary anterior teeth.
(Courtesy: Dr. Robert Margeas.)

deeper tooth preparation that is often located in dentin. Studies show that bond strengths to dentin decline over time and that porcelain veneers placed in intra-enamel preparations offer the best long-term results.[34–37] However, newer, currently available pressed or castable ceramics are capable of being fabricated to much thinner dimensions, making them viable options for indirect fabrication as well.

Although two appointments are required for indirect veneers, chair time is reduced because much of the work has been done in the laboratory. Excellent results can be obtained when proper clinical evaluation and careful operating procedures are followed. Indirect veneers are attached to the enamel by acid etching and bonding with light-cured resin cement.

I. No-Prep Veneers

As noted earlier (see section Veneers), one approach being used for indirect veneers is to place them on teeth with no tooth preparation. Although this 'no-prep' approach may at first appear desirable, it can later cause problems if proper case selection is not done.

No-prep veneers are best indicated when:

I. teeth are inherently undercontoured,
II. when interdental spaces or open incisal embrasures are present or
III. when both conditions exist.

Examples of successful no-prep veneers following these guidelines are seen in Figs. 21.37 and 21.38.

However, no-prep veneers can be *problematic*:

I. First, no-prep veneers are inherently made thinner and consequently are more prone to fracture, especially during the try-in phase.
II. Second, for indirect no-prep veneers, interproximal areas are difficult to access for proper finishing.
III. And third, as noted earlier, if case selection is not done properly and the teeth are already of normal contour, the resulting veneers inevitably will be over-contoured. Veneers that are over-contoured are not generally esthetic and often can result in impingement of gingival tissue, as noted earlier (Fig. 21.32D).



Figure 21.37 No-prep veneers placed on maxillary anterior teeth. (A) Before treatment and (B) immediately after placement of the no-prep veneers.
(Courtesy: Dr. Patricia Pereira.)

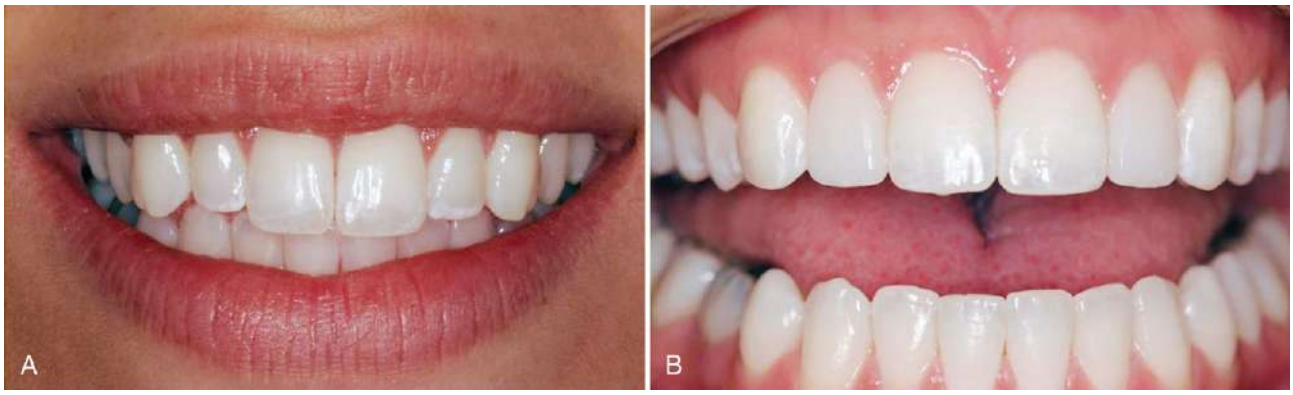


Figure 21.38 No-prep veneers placed to restore undersized maxillary lateral incisors. (A) Before treatment and (B) after placement of the no-prep veneers.
(Courtesy: Dr. Gary Radz.)

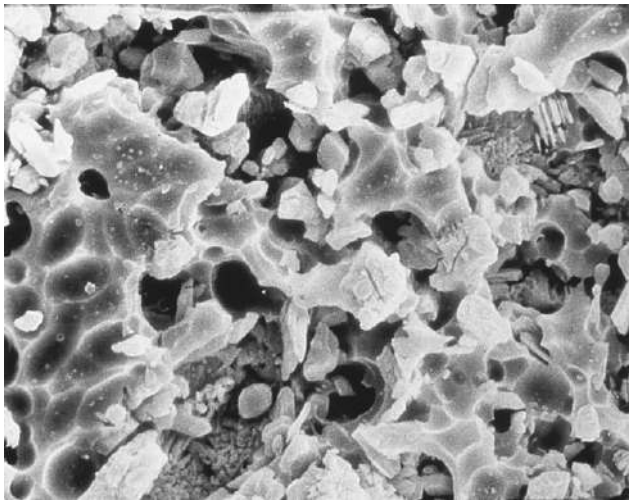


Figure 21.39 Scanning electron micrograph (×31,000) of feldspathic porcelain etched with hydrofluoric acid.
(Courtesy: Dr. Steven Bayne.)

For these reasons, it is advisable in most cases to use a conservative, intra-enamel preparation for the use of indirect veneers, as noted in the next section.

II. Etched Porcelain (Feldspathic) Veneers

The preferred type of indirect veneer is the etched porcelain (i.e. feldspathic) veneer.

- Porcelain veneers etched with hydrofluoric acid are capable of achieving high bond strengths to the etched enamel via a resin-bonding medium.[38–40] This porcelain etching pattern is shown in Fig. 21.39.
- In addition to the high bond strengths, etched porcelain veneers are highly esthetic, stain resistant and periodontally compatible. The incidence of cohesive fracture for etched porcelain veneers is also very low.[35]
- However, as noted earlier, the key to the long-term success with etched porcelain veneers is the use of a *conservative intra-enamel preparation*.
- Preparations into dentin should be avoided because virtually all problems associated with etched porcelain veneers (debonding, accelerated marginal staining, tooth sensitivity etc.) occur when excessive amounts of dentin are exposed in the veneer preparation.

Tooth Preparation

Because the most important consideration in determining the success of etched porcelain veneers is tooth preparation, a systematic approach will be presented first, using a dentiform series, prior to reviewing the associated clinical procedures (Fig. 21.40). The tooth used to illustrate the preparation sequence has intentionally been coloured blue to allow better visualization of the involved steps. However, it is not recommended that this tooth marking be done clinically.

A veneer design using a butt-joint incisal edge

- The veneer preparation is made with a *tapered, rounded-end diamond instrument*. It is critical that the tip diameter of the diamond be measured because the diamond will serve as the measuring tool in gauging proper reduction depth. A diamond with a *tip diameter of 1–1.2 mm is recommended*. The tip diameter of the diamond used in this series is 1.2 mm.
- The first step in the veneer preparation is establishing the *peripheral outline form*. Position the diamond to half its depth (approximately 0.6 mm in this example) just facial to the proximal contact on either proximal surface, and then extend the bur, while maintaining its occlusogingival orientation, around the gingival area and then back up the opposite proximal area, again keeping the diamond positioned just facial to the proximal contact area (Fig. 21.40A and B).
- In this example, a *supra-gingival marginal position* was maintained. Recall that clinically, the position of the gingival margin is determined by lip position (and the resulting display of teeth) and the gingival extent of the facial discolouration or defect being treated, as noted previously. If possible, a supra-gingival margin position is always considered desirable because it minimizes the potential for an adverse gingival response.
- ***Facial reduction*** is achieved by first identifying and then reducing three separate facial zones (Fig. 21.40C), in that order:
  - *Incisal third*. To prepare the incisal zone of facial reduction, the diamond first is aligned parallel with the facial surface of the incisal third of the tooth. The diamond is then moved mesio-distally from line angle to line angle until the desired depth of approximately 0.6 mm is attained. Again, the tip of the diamond is used to gauge this reduction. Reduction depth can be verified by viewing the tip of the diamond in proximity to the unprepared tooth structure gingival to this reduced area when viewed from the proximal, facial and incisal aspects (Fig. 21.40D and E). Care also must be taken to round the mesial and distal facial line angles during this reduction sequence to ensure uniform facial reduction.
  - *Middle third*. The middle third is reduced in a similar manner. By carefully watching the striations being created by the diamond mesio-distally during the reduction of the middle third, it is easy to see when the level of the previous incisal third reduction is reached (Fig. 21.40F). When a similar reduction level has been reached, the striations in the middle one-third will then extend into the area previously reduced in the incisal third. Stop immediately. *Do not go deeper.* Again, a reduction depth of approximately 0.6 mm is desirable. Moreover, the reduction depth again can be verified by viewing the tip of the

diamond in proximity to the unprepared tooth structure gingival to this reduced area when viewed from the proximal, facial and incisal aspects. Care also must be taken to round the mesial and distal facial line angles during this reduction sequence to ensure uniform facial reduction.

- *Gingival third*. Reduction of the gingival one third is straightforward and simply involves removal of the remaining 'island' of unprepared tooth structure to a level consistent with the surrounding previously prepared tooth structure (Fig. 21.40G and H).

- ***Incisal reduction*** is made by orienting the diamond perpendicular to the incisal edge and then reducing the incisal edge to attain a *minimum* reduction of 1 mm or, more desirably, 1.5 mm (Fig. 21.40I). Clinically, this reduction in depth will be gauged using an incisal

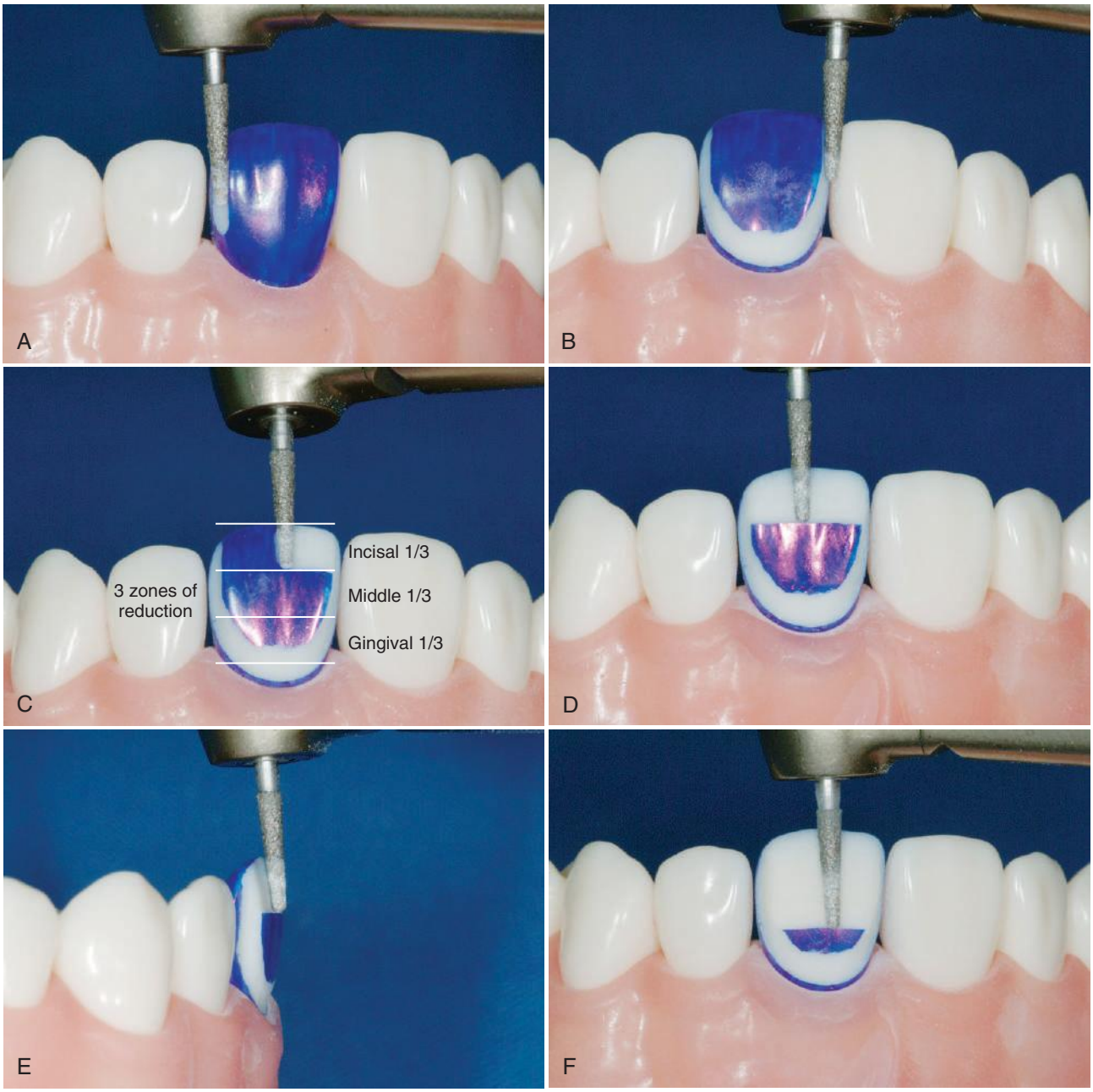


Figure 21.40 Intra-enamel preparation for an etched porcelain veneer with a butt-joint incisal edge design. (A and B) The peripheral outline form is first established using a rounded-end diamond instrument; (C–H) facial reduction is achieved by first identifying and then reducing three separate facial zones: the incisal third, the middle third and the gingival third (C), in that order;

(*Continued*)

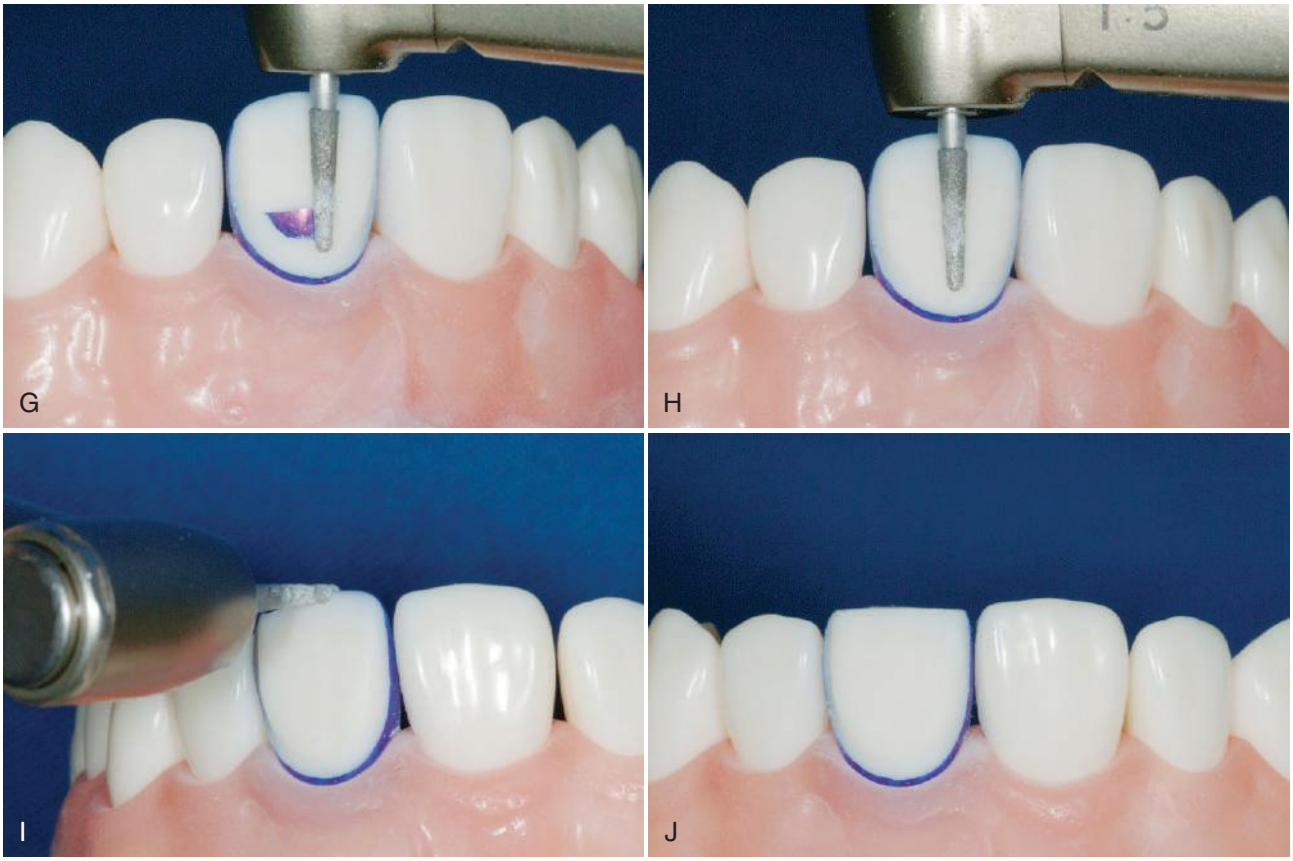


Figure 21.40 (*cont.*) (I) incisal reduction is attained; (J) the completed intra-enamel preparation for an etched porcelain veneer with a butt-joint incisal edge.

reduction index. In this dentiform example, a minimum 1-mm incisal reduction depth was generated. Finally, round the facioincisal line angle with the side of the diamond to reduce internal stresses in the porcelain veneer. The final intra-enamel preparation for an etched porcelain veneer using a butt-joint incisal edge design is seen in Fig. 21.40J.

A veneer design using incisal-lapping preparation

Frequently, an incisal-lapping preparation is indicated:

I. If the patient has worn or defective areas on the lingual aspect of the incisal edge.
II. Some operators also prefer this design because of enhanced adaptation of the veneer to the lingual preparation margin attributable to a 'lap sliding' fit.

The preparation steps for the incisal-lapping preparation are identical to those for the butt-joint design, including the steps for incisal reduction; however, additional steps are required to attain the incisal-lapping feature.

- The first step in achieving this preparation design is to notch the mesial and distal incisal angles. The tip of the same diamond instrument used for the earlier steps of the veneer preparation is used to establish these notches.
- Using the diamond, extend the notches completely through the incisal angles faciolingually to a depth incisogingivally consistent with the desired amount of lapping of the lingual surface (Fig. 21.41A). For example, if a 0.5-mm lap of the lingual surface is desired, as in this example, the notches are prepared to a depth of 0.5 mm each accordingly (Fig. 21.41B).
- Once the incisal notches have been generated incisogingivally to a depth consistent with the desired amount of lingual lapping, the preparation of the lingual lap is made. Position the diamond into the tooth to a depth of approximately 0.6 mm (less if remaining faciolingual thickness of the incisal edge enamel is compromised) and extend the preparation across the lingual surface from notch to notch (Fig. 21.41C).
- The resulting sharp incisal angles must then be rounded to finish the incisal-lapping portion of the preparation. Care must be taken to include any desired lingual defect.
- The gingival extent of the incisal lap is determined by the extent of any lingual defect.
- The final lapping portion of the preparation is seen in Fig. 21.41D. The facial view of the completed incisal lapping preparation with a lingual lap of 0.5 mm is seen in Fig. 21.41E.

Clinical Procedures

The case of a patient with generalized fluorosis is used to illustrate the clinical procedures involved (Fig. 21.42A).



Figure 21.41 Intra-enamel preparation steps for an etched porcelain veneer with an incisal-lapping design. (A and B) Incisal notches; (C) preparing the lingual-lapping portion of the prep and (D and E) the completed intra-enamel preparation viewed from the lingual and facial aspects for an etched porcelain veneer with an incisal-lapping design.

Preoperative clinical considerations

A consult appointment is always recommended prior to initiating the veneer procedures. At this appointment, the actual procedures are discussed in detail, appropriate consents are obtained, and any needed records are generated, including shade selection, intra-oral photographs, and impressions for diagnostic models and occlusal records. Laboratory communications are greatly enhanced through the inclusion of digital photographs. An excellent series of baseline photographs, including some with shade tabs positioned in the photographic field to document the preoperative shapes and shades of the involved teeth, should be made.

Tooth preparation protocol for veneers

- Although *intra-enamel preparations* will be used, it is always recommended that patients be anaesthetized during the appointment for tooth preparation to ensure maximum comfort for the patient and the dentist. Once the anesthetic is administered, preoperative records such as an incisal reduction index and those needed to facilitate temporization are made.
- An *incisal reduction index* is always recommended to accurately gauge the amount of incisal reduction during the preparation of teeth for etched porcelain veneers (Fig. 21.43; see also Fig. 21.42B–H). An incisal



Figure 21.42 Etched porcelain veneers using an intra-enamel preparation. (A) A patient with severe dental fluorosis; (B) an incisal reduction index is made intra-orally, since no significant change in incisal edge position is desired; (C) retraction cord is placed. (D) The outline form is first established. (E–G) Facial reduction is attained by using three zones of facial reduction. (H) Incisal reduction is verified using the incisal reduction index.

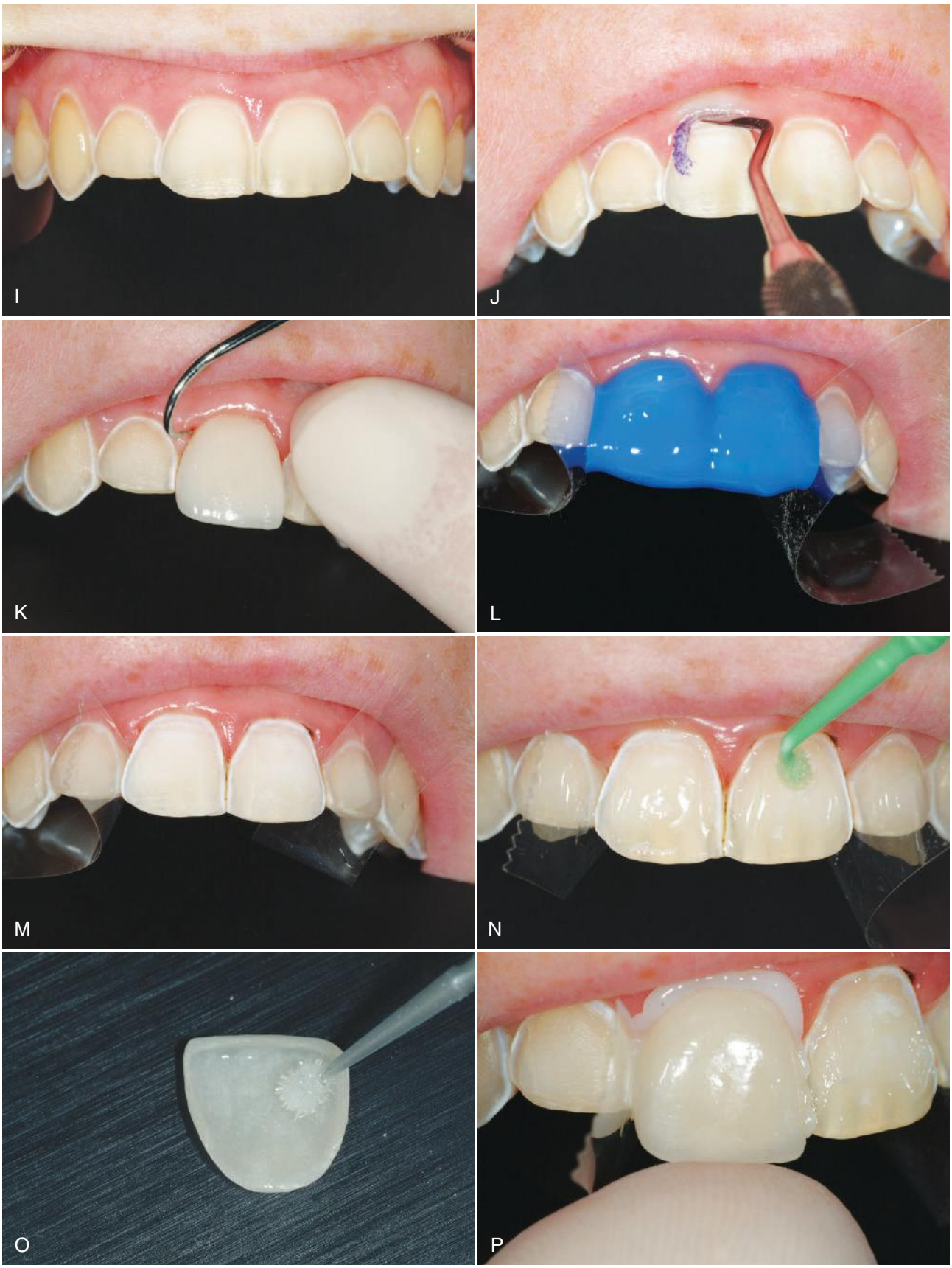


Figure 21.42 (*cont.*) (I) Finished preparations for intra-enamel preparations. Note the window preparations on canines and premolars. (J) Retraction cord is placed for isolation. (K) The fit of the veneer is assessed. (L and M) Etching of the prepared maxillary central incisors. (N and O) Adhesive is applied to the etched enamel and the tooth side of the porcelain veneer. (P) The veneer is loaded with resin cement and seated on the tooth.

(*Continued*)



Figure 21.42 (*cont.*) (Q) Excess cement is removed with a microbrush. (R) Excess cement is removed interproximally through removal of polyester strip. (S) Resin cement cured with intense curing light. (T) No. 12 surgical blade in a Bard-Parker handle is used for removing excess cured resin cement. (U and V) Diamond instruments used to 'dress' marginal areas. (W and X) 30-Fluted carbide burs and diamond impregnated polishing instruments used to finish and polish veneer margins.



Figure 21.42 (*cont.*) (Y and Z) Finished etched porcelain veneers as viewed from the lingual and facial aspects.

reduction index is made by recording the lingual and incisal contours of the anterior teeth to be prepared or the contours generated in a diagnostic wax-up. Typically, a fast-setting silicone or polyvinyl siloxane (PVS) elastomeric material is used to generate this record. If a diagnostic model has been made in which the incisal edge positions of the final veneers will be different from the current teeth, the incisal reduction index should be made using the diagnostic model (Fig. 21.43A and B). As in this case, if no change in the incisal edge positions of the involved teeth is desired, the reduction index can be made directly in the patient's mouth by recording the existing contours of the involved teeth. Once the facial excess has been trimmed away with a No. 12 surgical blade in a Bard-Parker handle, the incisal reduction index should be tried in the mouth to verify the accuracy of the index (Fig. 21.42B).



Figure 21.43 Incisal reduction index made from a diagnostic model. (A and B) A fast-set elastomeric material is used to record the lingual and incisal contours of the diagnostic model; (C) incisal reduction index is used to verify proper incisal preparation of teeth and (D) finished etched porcelain veneers.

- The patient in this case has a high smile line and generalized defects that involve the entire facial surfaces of anterior teeth. Therefore, a small diameter gingival retraction cord is placed in anticipation of veneer margins that will be placed at the level of the free gingival margins (Fig. 21.42C).
- *Incisal lapping preparations* for porcelain veneers are made on the maxillary central and lateral incisors consistent with the systematic step-by-step preparation procedures described earlier. The intra-enamel preparations are made with a tapered, rounded-end diamond instrument to a depth of approximately 0.5–0.75 mm midfacially, diminishing to a depth of 0.3–0.5 mm along the gingival margins, depending on enamel thickness. The outline form is first established as noted earlier (Fig. 21.42D). Facial reduction is attained by using three zones of facial reduction as previously described (Fig. 21.42E–G).
- Veneer interproximal margins *should extend* into the facial and gingival embrasures, without engaging an undercut, and yet should be located just facial to the proximal contacts.
- Since this patient is young with beautifully shaped canines and also exhibits a canine-guided occlusion, a *window type of preparation design* was employed when preparing the maxillary canines. By restricting the window prep veneers to the facial surface entirely, functional contact will be maintained solely on tooth structure, thereby preventing accelerated wear of opposing teeth. Adequate incisal reduction is verified using the incisal reduction index as noted earlier (Fig. 21.42H). The finished preparations are shown in Fig. 21.42I.
- After the preparations are completed, an *elastomeric impression* is made and appropriate occlusal records generated.
- *Digital photos* of the prepared teeth (with and without appropriate shade tabs in the photographic field) also are recommended.
- In most cases, *provisionals* are fabricated for the prepared teeth, as described in the subsequent section on temporization (see the Veneer Provisionalization section). If a diagnostic wax-up was generated, veneer provisionals are necessary to assess the tooth contours generated in the wax-up intra-orally. If no diagnostic wax-up was needed, occasionally temporary restorations are not required if the patient consents, because the preparations are shallow and involve only enamel. In these cases, the patient should be instructed to avoid biting hard foods and objects, to keep the areas clean with a soft bristled brush and to expect the possibility of some mild sensitivity to hot and cold. It should be noted that the vast majority of veneer cases will warrant temporization.

Veneer cementation protocol

- After they have been fabricated in the laboratory, the porcelain veneers are returned to the dentist for cementation at the second appointment. The completed veneers must be *inspected* for cracks, overextended margins and adequate internal etching (as evidenced by a frosted appearance). Marginal areas in particular should be inspected for proper etching so that an adequate seal occurs in these areas. Overextended marginal areas inter-proximally may preclude full seating of adjacent veneers. These areas can be trimmed carefully with a micron-finishing diamond instrument or flexible abrasive disk. Unless severe or inaccessible, most minor overextensions should be trimmed only after bonding the veneer to the tooth because of the risk of fracturing the porcelain.
- After the prepared teeth are cleaned with a pumice slurry, rinsed and dried, *isolation* is accomplished with a lip retractor (optional) and cotton rolls. A 5-cm-by-5-cm (2-in-by-2-in) cotton gauze is placed across the back of the patient's mouth to protect against aspiration or swallowing of an inadvertently dropped veneer.
- If the veneer margins closely approximate the gingiva, a small diameter *retraction cord* should be placed in the gingival crevice during try-in and cementation to prevent inadvertent contamination of the bonds from crevicular fluids (Fig. 21.42J).
- The fit of each veneer is evaluated on the respective individual tooth and adjusted if necessary. A No. 2 explorer should be used to assess marginal fit (Fig. 21.42K). All of the veneers should be tried in place not only individually but also in adjacent pairs to ensure the fit of adjacent seated veneers.
- Veneers should be tried in place only on clean, dry teeth to eliminate any potential for contamination. If accidental contamination occurs, the veneer should be cleaned thoroughly with alcohol or acid etchant, rinsed and dried before bonding.
- Prior to cementation, a silane agent can optionally be applied to the internal surfaces of the veneers. The silane acts as a coupling agent, forming a chemical bond between the porcelain and the resin that increases bond strength of the resin to the porcelain.[41] It also improves the wettability of the porcelain. *The primary source of retention with porcelain veneers still remains the etched porcelain surface itself.* Only a modest increase in bond strength results from silanation of the porcelain; however, it is recommended because it also may reduce marginal leakage and discolouration.
- A *light-cured resin cement* is recommended for bonding the veneer to the tooth because light-cured resins are more colour stable and provide additional working time over the self-cured or dual-cured versions. Shade selection of the bonding medium is determined after the fit of the individual veneers has been evaluated and confirmed. Water can be used as an optical medium for try-in to assess the appearance of the veneers. With this technique, water is placed in a single central

incisor veneer (or both central incisor veneers); placed on clean, dry teeth; and the appearance assessed. If the veneers are deemed acceptable in appearance, an *untinted shade* of the bonding resin is indicated. For the vast majority of etched porcelain veneers, an untinted shade of the bonding resin is recommended. Because etched porcelain veneers can be fabricated with inherent colour gradients, characterizations and even additional opaques when further masking of stains is needed, it is unusual that alternative shades or opaque resin bonding media are needed. Nonetheless, alternative shades or opaque versions of resin bonding cements are sometimes preferred by some dentists.

- If alternative shades of resin cements are needed, *shade selection* is made by first placing a uniform layer of a selected shade of resin cement, approximately 0.5 mm in thickness, on the tooth side of a single veneer. Typically, a central incisor veneer is used to facilitate shade determination of the cement. The operatory light is turned away during shade assessment to prevent premature and inadvertent curing of the resin cement. The veneer is seated on a clean, dry, unetched tooth; the excess resin cement is removed with a brush; and the overall shade of the veneer is evaluated. After try-in, the veneer is removed quickly and stored in a container that is impervious to light to prevent curing of the cement. If the shade of the cement is determined to be appropriate, more of the same shade is added to the veneer just before bonding. If a different shade is deemed necessary, the existing shade is wiped from the inner aspect of the veneer with a disposable microbrush, and a new shade of resin cement is placed in the veneer. In the meantime, the assistant can remove residual cement of the previous shade from the tooth with a cotton pellet or brush. The veneer loaded with the new shade of cement is reseated and evaluated, as previously described.
- Water-soluble *try-in pastes* that correspond to the same shades of resin cements also are available with many veneer-bonding kits. These try-in pastes allow shade assessment without the risk of inadvertent premature curing of the cement because the try-in pastes are not capable of setting. However, after the try-in process with these try-in pastes, the veneers must be thoroughly cleaned, according to the manufacturer's instructions, to ensure that optimal bonding occurs. A light-cured resin cement of the same shade is used for final cementation.
- The inherent shade of the veneer, characterization and internal opaquing must be accomplished primarily during the fabrication of the veneer itself in the laboratory. Some additional masking can be accomplished chairside using an opaque resin-bonding medium at the time of bonding. However, excess use of opaque bonding resins also can reduce the 'esthetic vitality' of the veneer, resulting in a poor esthetic outcome. Also, the overall shade of the veneer can be modified only slightly by the shade of cement selected. Significant changes in shading or masking ability cannot be accomplished chairside.
- Prior to cementation of the veneers, the retraction cords are evaluated to ensure that they are tucked adequately into the gingival crevice. The tooth used for try-in to assess the shade of the resin-bonding medium should be cleaned again with a slurry of pumice to remove any residual resin or try-in paste that may preclude proper acid etching of the enamel.
- A technique recommended for applying the veneers one at a time is presented here. *Polyester strips* are placed interproximally to prevent inadvertent bonding to the adjacent tooth, followed by etching, rinsing and drying procedures. It is recommended that the two central incisors be etched and their veneers bonded first because of their critical importance esthetically. Etch the teeth to be bonded with a 35%–37% phosphoric acid gel (Fig. 21.42L). Following etching, rinsing and drying, the preparations should exhibit a frosted appearance as evidence of an appropriately etched enamel surface (Fig. 21.42M).
- An adhesive is applied to the etched enamel and the tooth side of the silane-primed porcelain veneer (Fig. 21.42N and O). Next, a thin layer (0.5 mm) of the selected shade of light-cured resin cement is placed on the tooth side of the veneer, taking care not to entrap air. The first veneer is placed on the tooth and vibrated (carefully and lightly) into position with a blunt instrument or light finger pressure (Fig. 21.42P). The margins of the veneer are examined with a No. 2 explorer to verify accurate seating. Next, the excess resin cement is removed with a disposable microbrush, always directing the microbrush in a gingival direction to prevent displacement of the veneer (Fig. 21.42Q). The second veneer is placed and cleaned of excess cement in like manner. If a veneer cement with thick viscosity is used, the polyester strips can be carefully removed to facilitate removal of interproximal resin prior to curing (Fig. 21.42R). However, the resin cement should be cured only after visual inspection reveals no excess resin remains in these critical interproximal areas.
- Veneer margins are evaluated again before the veneer is exposed to the curing light. To ensure complete polymerization, the veneer should be cured for a minimum of 20–40 s each from the facial and lingual directions with a high-intensity, blue light emitting diode light (Fig. 21.42S). A light stream of air can be directed on the tooth during the curing sequence to prevent overheating from the curing light. After positioning and bonding of the first two veneers on the central incisors, the remaining veneers can be positioned carefully and bonded in like manner. Following proper positioning and bonding of all the veneers, any residual resin cement can be removed. A No. 12 surgical blade

in a Bard-Parker handle is ideal for removing excess cured resin cement remaining around the margins (Fig. 21.42T). Care must be exercised to ensure that the surgical blade is used only with a secure finger rest and using short shaving strokes, always directed parallel to the veneer margins. Removal of the retraction cord at this time facilitates access and visibility to the sub-gingival areas. If the marginal fit of the porcelain veneers is deemed acceptable and a favourable emergence profile exists, only removal of the excess cement is required.

- If the porcelain margins are overextended beyond the cavosurface angles, an overhang is present or the marginal areas are too bulbous, recontouring of these areas is required (especially along the gingival margins) to ensure proper physiologic contours and gingival health. A *flame-shaped fine diamond instrument* is used to carefully recontour and 'dress' these areas (Fig. 21.42U). Marginal areas should be confluent with the surrounding unprepared tooth surfaces when assessed with a No. 2 explorer. The lingual areas are always finished with an oval-shaped fine diamond instrument (Fig. 21.42V). Because the use of a diamond instrument breaks the glazed surface, a series of appropriate instruments is used to restore a smooth surface texture.
  - First, a rounded-end or bullet-shaped (or oval for lingual surfaces), *30-fluted carbide finishing bur* (Midwest No. 9803 or Brasseler No. 7801) is used to plane the porcelain surface and to remove the striations created by the diamond instruments (Fig. 21.42W). Studies show that the best results occur if the diamond instruments are used with air and water coolant, whereas the 30-fluted bur should be used dry.[42]
  - Second, the porcelain is smoothed and polished with a series of *abrasive rubber, porcelain polishing cups* and *points* (Dialite Porcelain Polishing Kit; Brasseler USA, Savannah, GA) (Fig. 21.42X). Final surface lustre is imparted by using a porcelain-polishing paste, applied with either a rubber prophy cup or a felt wheel. This step is optional if a suitable polish has been attained with the polishing points and cups. The completed veneers are shown from incisal and facial views in Fig. 21.42Y and Z.

Etched porcelain veneers also can be used effectively to restore malformed anterior teeth conservatively. Malformed lateral incisors are shown in Fig. 21.44A. Incisal, lapping preparations that are extended well onto the lingual surface are used (Fig. 21.44B). The resulting restorations are virtually comparable with 'three-quarter crowns' in porcelain. The final esthetic results are shown in Fig. 21.44C.

Clinical considerations for darkly discoloured teeth

Darkly discoloured teeth are more difficult to treat with porcelain veneers. Several modifications in the veneering technique can be used to enhance the final esthetic result.

- First, opaque porcelain can be incorporated during the fabrication of the veneers to achieve more inherent masking. If the veneers are not inherently opaque, little chance exists for adequate masking of a darkly stained tooth. Typically, 5%–15% opaque porcelain is required to achieve optimal masking. Exceeding 15% opaque porcelain dramatically reduces light penetration and results in a significant loss of esthetic vitality; the esthetic vitality or the realistic appearance of teeth depends on light penetration (see section Translucency, under Artistic Elements).
- Second, a slightly deeper tooth preparation can be used to allow greater veneer thickness. The preparation should always be restricted to enamel, however, to ensure optimal bonding of the veneer to the tooth. Even with improved dentin-bonding agents, the bonds to dentin are less predictable or durable than the bonds to enamel because of the high variability and dynamic nature of dentin. Bonds to etched enamel are highly predictable and very durable.
- Third, the laboratory can be instructed to use several coats of a die-spacing medium on the laboratory model to allow a slightly greater thickness of the resin-bonding medium. The die-spacing medium must not be extended closer than 1 mm to the margins to ensure adequate positioning of the veneer to the preparation during try-in and bonding and to provide for a slight



Figure 21.44 Treatment of malformed teeth with porcelain veneers. (A) Malformed lateral incisors; (B) an incisal-lapping preparation similar to a three-quarter crown in enamel is used and (C) final esthetic results.

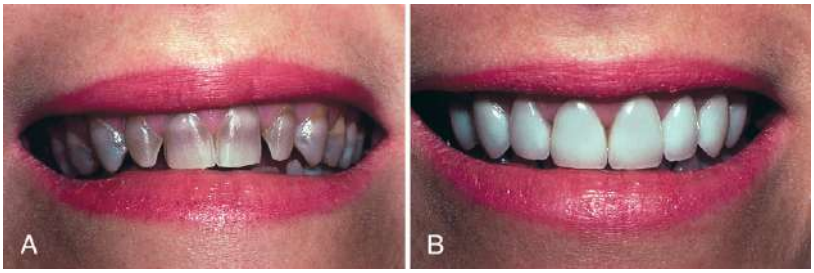


Figure 21.45 Darkly stained teeth treated with porcelain veneers. (A) Tetracycline-stained teeth seen after preparation for porcelain veneers and (B) after view of completed veneers.

internal space. A typical case of darkly discoloured teeth showing prepared teeth and postoperative result is shown in Fig. 21.45.

Patients who have darkly stained teeth always should be informed that although porcelain (or composite) veneers can result in improved esthetics, they may not entirely eliminate or mask extremely dark stains. Because of the limited thickness of the veneers and the absolute necessity of incorporating intrinsic opacity, the realistic translucency or esthetic vitality of veneered teeth may never be comparable with that of natural, unaffected teeth (Fig. 21.9). Full porcelain coverage with all-ceramic crowns may be indicated in some patients with severe discolouration because of the crown's greater capacity to restore esthetic vitality. Nonetheless, porcelain veneers are a viable option in most cases for patients who desire esthetic improvement without significant tooth reduction.

III. Pressed Ceramic Veneers

Another esthetic alternative for veneering teeth is the use of *pressed ceramics* (e.g. IPS Empress or e.max [Ivoclar Vivadent]). In contrast to etched porcelain veneers that are fabricated by stacking and firing feldspathic porcelain, pressed ceramic veneers are literally cast using a lost wax technique.

- *Excellent esthetics* is possible using pressed ceramic materials for most cases involving mild to moderate discolouration.
- Because of the more translucent nature of pressed ceramic veneers, however, *dark discolourations are best treated with etched porcelain veneers*.
- The clinical technique for placing pressed ceramic veneers (e.g. IPS Empress) is not markedly different from that for feldspathic porcelain veneers, other than the need for a *slightly greater tooth reduction depth*.
- The procedures for tooth preparation, try-in and bonding of pressed veneers are the same as for etched porcelain veneers except that the marginal fit is superior. For that reason, often little marginal finishing is necessary. Only the excess bonding medium needs to be removed. A typical case involving pressed ceramic veneers, with before and after treatment views, is shown in Fig. 21.46.



Figure 21.46 Pressed ceramic veneers (IPS Empress). (A–C) Before treatment, facial views and (D–F) esthetic result after completed veneers.
(Courtesy: Dr. Luiz N. Baratieri.)

Veneer Provisionalization

Because intra-enamel preparations for etched porcelain veneers are by design very conservative, the resulting provisionals are inherently thin. Furthermore, they cannot be bonded in a similar manner to conventional provisionals for crowns, for example, because of the lack of inherent retention form. Moreover, since they are very thin, they cannot be made in the mouth and removed for trimming and subsequent cementing because of the high probability of fracture. Therefore, veneer provisionals must be made and placed simultaneously intra-orally.

Fig. 21.47 illustrates a typical case involving the fabrication of provisionals for etched porcelain veneers for maxillary anterior teeth. A clear PVS material is used to make the preoperative impression from which the provisionals will be made. If no diagnostic model is needed and the existing contours of the teeth are to be replicated, the impression for the provisionals is made directly in the mouth. However, as in this case, if a diagnostic wax-up was generated (Fig. 21.47A and B), the clear PVS impression is made from this diagnostic model. Making the temporary from the diagnostic wax-up enables the clinician to see the contours of the wax-up manifested intra-orally in the resulting provisionals. As seen in Fig. 21.47C, the impression itself is removed from the outer tray (no tray adhesive is used) and set aside for future use.

- Following the preparation and impression of the teeth for porcelain veneers, the teeth to be temporized are *spot etched* with 35%–37% phosphoric acid.
- Only a *2-mm circle of enamel* should be etched on the facial surface of each tooth to be veneered (Fig. 21.47D and E).
- Because of the low viscosity of the bis-acryl temporary material, *no bonding agent is required for bonding*. The bis-acryl material will infiltrate the etched areas for micromechanical bonding. These spot-etched areas will be the only areas to which the veneer provisionals will be bonded. If the entire tooth were etched, the veneer provisionals could not be readily removed. After spot etching, only small-etched circles evidenced by a frosted appearance should be present on the surface of each prepared tooth (Fig. 21.47F).
- The etched teeth must be kept clean and dry at this point. The clear PVS impression is quickly loaded with a self-curing bis-acryl temporary material and thereafter is immediately positioned in the mouth (Fig. 21.47G and H).
- Once seated, finger pressure is applied to the peripheral areas of the flexible impression (since no outer hard tray is present) to express the excess material and 'thin out' the resulting resin 'flash'.
- When the bis-acryl temporary material has set, the clear impression is removed (Fig. 21.47I). The gross excess 'flash' material facially and lingually is removed with cotton pliers. Thereafter, a No. 12 blade held in a Bard-Parker surgical handle is used to *carefully* trim the excess temporary material around the margins of each tooth (Fig. 21.47J). The same No. 12 blade is used to carefully trim excess material in the gingival embrasure areas as well.
- A sharp large discoid applied parallel to the lingual margins is used to remove resin 'flash' in the lingual concave areas. The temporary veneers are all joined together interproximally, increasing their collective strength and enhancing retention.
- A light-cured resin glazing agent is applied and cured to generate a smooth surface texture (Fig. 21.47K). Final views of the finished provisionals are seen in Fig. 21.47L and M.
- Appropriate adjustments can then be made intra-orally in the provisionals to optimize occlusion and esthetics. A digital photograph of the final provisionals should be shared with the laboratory as a template for the final veneers. Patients must be instructed that they should not bite anything of any substance because of the weak nature of these veneers. These provisional veneers literally are more to accommodate esthetics and not function during the interim time until delivery of the final veneers.
- As demonstrated in a different case, once the patient returns for the final try-in and cementation of the veneers, the provisionals are carefully removed by prying them from each tooth using a Black's spoon (Fig. 21.47N). The provisionals can readily be removed since the only area where they actually are bonded to the tooth is the very small spot-etched, 2-mm circle on the facial surface of each prepared tooth.
- Once the veneer provisionals are removed, the areas that had been spot etched and bonded need to be lightly resurfaced with a flame-shaped diamond to ensure no residual resin-bonding agent is present that could preclude proper seating and bonding of the final veneers (Fig. 21.47O).

Veneers for Metal Restorations

Esthetic inserts (i.e. partial or full veneers) of a tooth-coloured material can be placed on the facial surface of a tooth previously restored with a metal restoration. For new castings, plans are made at the time of tooth preparation and during laboratory development of the wax pattern to incorporate a veneer into the cast restoration. After such a casting has been cemented, the veneer can be inserted, as described in the next section, except that the portion of mechanical retention of the veneer into the casting has been provided in the wax pattern stage.

Veneers for Existing Metal Restorations

Occasionally, the facial portion of an existing metal restoration (amalgam or gold) is judged to be distracting (Fig. 21.48A). A careful examination, including a

radiograph, is required to determine that the existing restoration is sound before an esthetic correction is made. The size of the unesthetic area determines the extent of the preparation. Anaesthesia is not usually required because the preparation is in metal and enamel. Preliminary procedures consist of cleaning the area with pumice, selecting the shade and isolating the site with a cotton roll. When the unaesthetic metal extends sub-gingivally, the level of the gingival tissue is marked on the restoration with a sharp explorer and a retraction cord is placed in the gingival crevice. Rubber dam isolation may be required in some instances.

A No. 2 carbide bur rotating at high speed with an air-water spray is used to remove the metal, starting at a point midway between the gingival and occlusal margins. The preparation is made perpendicular to the surface (a minimum of approximately 1 mm in depth), leaving a *butt joint* at the cavosurface margins. The 1-mm depth and butt joint should be maintained as the preparation is extended occlusally. All of the metal along the facial enamel is removed, and the preparation is extended into the facial and occlusal embrasures just enough for the veneer to hide the metal. The contact areas on the proximal or



Figure 21.47 Temporization for etched porcelain veneers. (A and B) Diagnostic models in anticipation of etched porcelain veneers; (C) clear polyvinyl siloxane (PVS) impression made from diagnostic model; (D–F) spot-etched areas for retention of temporary veneers;

(*Continued*)



Figure 21.47 (*cont.*) (G and H) Clear PVS impression is quickly loaded with bis-acryl temporary material, and positioned in the mouth; (I and J) impression is removed, and No. 12 blade in a Bard-Parker surgical handle is used to remove excess material; (K) glazing agent placed and cured.

(*Continued*)



Figure 21.47 (*cont.*) (L and M) final facial and lingual views of veneer provisionals and (N and O) at delivery appointment, temporary veneers are removed with Black's spoon and the spot-etched and bonded areas are relieved with a diamond.

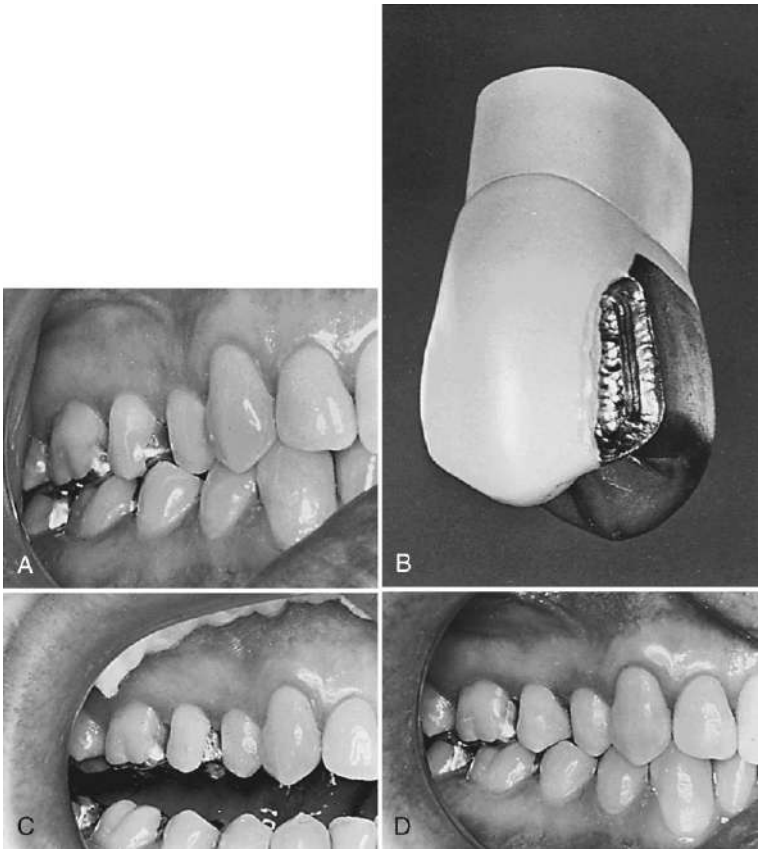


Figure 21.48 Veneer for existing cast restoration. (A) Mesiofacial portion of onlay is distracting to patient. (B) Model of tooth and preparation. Note 90° cavosurface angle and retention prepared in gold and the cavosurface bevel in enamel. (C) Clinical preparation ready for composite resin. (D) Completed restoration.

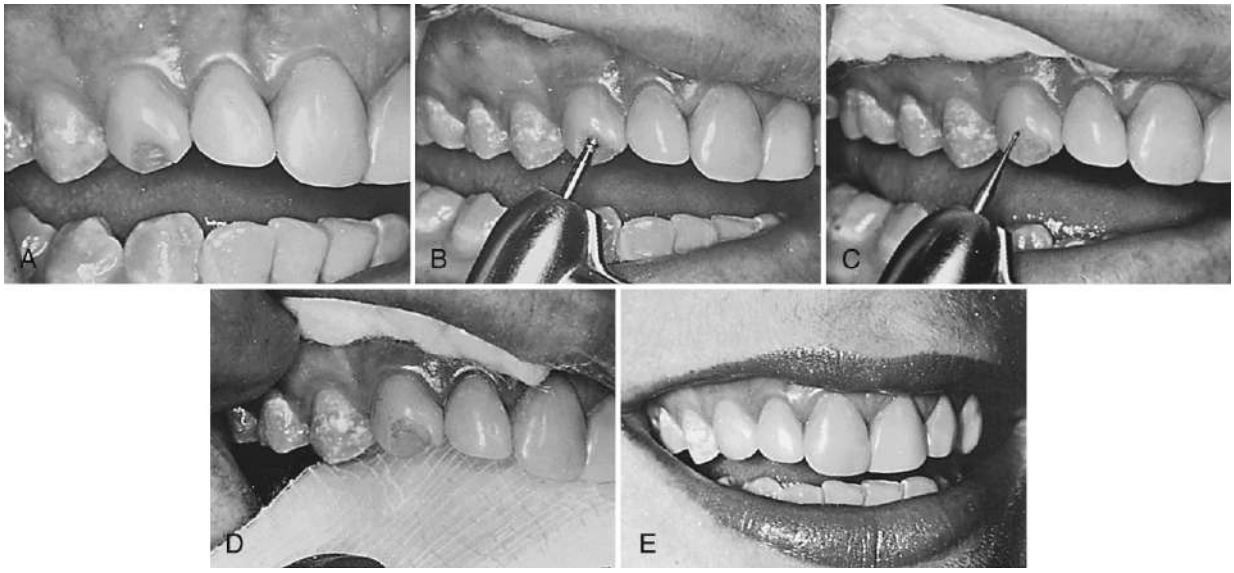


Figure 21.49 Repairing a direct composite veneer. (A) Fractured veneer on the maxillary canine; (B) preparation with rounded-end diamond instrument; (C) undercuts placed in existing veneer with a No. 1/4 bur; (D) completed preparation is shown isolated and etched and (E) veneer restored to original colour and contour.

occlusal surfaces must not be included in the preparation. To complete the outline form, the preparation is extended gingivally approximately 1 mm past the mark indicating the clinical level of the gingival tissue.

The final preparation should have the same features as those described for veneers in new cast restorations. Mechanical retention is placed in the gingival area with a No. 1/4 carbide bur (using air coolant to enhance vision) 0.25 mm deep along the gingivoaxial and linguoaxial angles. Retention and esthetics are enhanced by bevelling the enamel cavosurface margin (approximately 0.5 mm wide) with a coarse, flame-shaped diamond instrument oriented at 45° to the external tooth surface (Fig. 21.48B). After it is etched, rinsed and dried, the preparation is complete (Fig. 21.48C). Adhesive resin liners containing 4-methyloxy ethyl trimellitic anhydride, capable of bonding composite to metal, also may be used but are quite technique sensitive.[43] Manufacturers' instructions should be followed strictly to ensure optimal results with these materials. The composite material is inserted and finished in the usual manner (Fig. 21.48D).

Repairs of Veneers

Failures of esthetic veneers occur because of breakage, discolouration or wear. Consideration should be given to conservative repairs of veneers if the examination reveals that the remaining tooth and restoration are sound. It is not always necessary to remove all of the old restoration. The material most commonly used for making repairs is *light-cured composite*.

Small chipped areas on veneers often can be corrected by recontouring and polishing. When a sizable area is broken, it usually can be repaired if the remaining portion is sound (Fig. 21.49A). For direct composite veneers, repairs ideally should be made with the same material that was used originally. After cleaning the area and selecting the shade, the operator should roughen the damaged surface of the veneer or tooth (or both) with a coarse, tapered, rounded-end diamond instrument to form a *chamfered cavosurface margin* (Fig. 21.49B). Roughening with micro-etching (i.e. sandblasting) also is effective. For more positive retention, *mechanical locks* may be placed in the remaining composite material with a small, round bur (Fig. 21.49C). Acid etchant is applied to clean the prepared area and to etch any exposed enamel, which is then rinsed and dried (Fig. 21.49D). Next, an adhesive is applied to the preparation (i.e. existing composite and enamel) and polymerized. Composite is added, cured and finished in the usual manner (Fig. 21.49E).

To repair porcelain veneers, a hydrofluoric acid gel, suitable for intra-oral use (but only with a rubber dam in place), must be used to etch the fractured porcelain. Hydrofluoric acid gels are available in approximately 10% buffered concentrations that can be used for intra-oral porcelain repairs if proper isolation with a rubber dam is used. Although caution still must be taken when using hydrofluoric acid gels intra-orally, the lower acid concentration allows for relatively safe intra-oral use. Full-strength hydrofluoric acid should *never* be used intra-orally for etching porcelain. Isolation of the porcelain veneer to be repaired should always be accomplished with a rubber dam to protect gingival tissue from the irritating effects of the hydrofluoric acid. The manufacturer's instructions must be followed regarding application time for the hydrofluoric acid gel to ensure optimal porcelain etching. A lightly frosted appearance, similar to that of etched enamel, should be seen if the porcelain has been properly

etched. A silane coupling agent may be applied to the etched porcelain surface before the adhesive is applied. Composite material is added, cured and finished in the usual manner. Large fractures are best treated by replacing the entire porcelain veneer.

Summary

Conservative esthetic procedures help dentist to restore a wide range of esthetic challenges ranging from surface malformations to discolorations of varying complexities. An understanding of the various artistic elements in smile designing is important for a clinician to be able to satisfy the patient's expectations. The inherent advantages and limitations of each of these procedures—correction of diastemata, bleaching treatments, microabrasion and macro-abrasion to veneers are to be taken into consideration in order to realize their optimal clinical performance.

Acknowledgements

Portions of the section on artistic elements were reprinted with permission from Heymann HO: The artistry of conservative esthetic dentistry, *J Am Dent Assoc* 115 (12E; special issue):14, 1987.

References

1. Goldstein RE: *Esthetics in dentistry*, Philadelphia, PA, 1976, JB Lippincott.
2. Levin EI: Dental esthetics and the golden proportion, *J Prosthet Dent* 40:244, 1978.
3. Borissavlievitch M: *The golden number*, London, 1964, Alec Tiranti.
4. Preston JD: The golden proportion revisited, *J Esthet Dent* 5:247–251, 1993.
5. Ward DH: A study of dentists' preferred maxillary anterior tooth width proportions: comparing recurring esthetic dental proportions to other mathematical and naturally occurring proportions, *J Esthet Rest Dent* 19(6):324–339, 2007.
6. Sterrett JD, Oliver T, Robinson F, et al: Width/length ratios of normal clinical crowns of maxillary anterior teeth, *J Clin Periodontol* 26:153–157, 1999.
7. Magne P, Gallucci GO, Belser UC: Anatomic crown width/length ratios of unworn and worn maxillary teeth in white subjects, *J Prosthet Dent* 89:453–461, 2003.
8. Chiche GJ, Pinault A: *Esthetics of anterior fixed prosthodontics*, Chicago, 1994, Quintessence.
9. Sproull RC: Understanding color. In Goldstein RE, editor: *Esthetics in dentistry*, Philadelphia, PA, 1976, JB Lippincott.
10. Dawson PE: *Evaluation, diagnosis, and treatment of occlusal problems*, ed 2, St. Louis, MI, 1989, Mosby.
11. Graber TM, Vanarsdall RL: *Orthodontics: current principles and techniques*, ed 3, St. Louis, MI, 2000, Mosby.
12. Hattab FN, Qudeimat MA, al-Rimawi HS: Dental discolorations: an overview, *J Esthet Dent* 11:291, 1999.
13. Carver CC, Heymann HO: Dental and oral discolorations associated with minocycline and other tetracycline analogs, *J Esthet Dent* 11:43, 1999.
14. Haywood VB: History, safety, and effectiveness of current bleaching techniques and applications of the nightguard vital bleaching technique, *Quintessence Int* 23:471, 1992.
15. Haywood VB: Nightguard vital bleaching: a history and products update: Part 1, *Esthet Dent Update* 2:63, 1991.
16. Titley KC, Torneck CD, Ruse ND: The effect of carbamide-peroxide gel on the shear bond strength of a microfill resin to bovine enamel, *J Dent Res* 71: 20, 1992.
17. Sorensen JA, Martinoff JT: Intracoronal reinforcement and coronal coverage: a study of endodontically treated teeth, *J Prosthet Dent* 51:780, 1984.
18. Harrington GW, Natkin E: External resorption associated with bleaching of pulpless teeth, *J Endod* 5:344, 1979.
19. Holmstrup G, Palm AM, Lambjerg-Hansen H: Bleaching of discolored root-filled teeth, *Endod Dent Traumatol* 4:197, 1988.
20. Lado EA: Bleaching of endodontically treated teeth: an update on cervical resorption, *Gen Dent* 36:500, 1988.
21. Madison S, Walton R: Cervical root resorption following bleaching of endodontically treated teeth, *J Endod* 16:570, 1990.
22. Haywood VB, Heymann HO: Nightguard vital bleaching: how safe is it? *Quintessence Int* 22:515, 1991.
23. Lado EA, Stanley HR, Weisman MI: Cervical resorption in bleached teeth, *Oral Surg* 55:78, 1983.
24. Feinman RA, Goldstein RL, Garber DA: *Bleaching teeth*, Chicago, IL, 1987, Quintessence.
25. Haywood VB, Heymann HO: Nightguard vital bleaching, *Quintessence Int* 20:173, 1989.
26. Sagel PA, Odioso LL, McMillan DA, et al: Vital tooth whitening with a novel hydrogen peroxide strip system: design, kinetics, and clinical response, *Compend Cont Educ Dent* 29(Suppl):S10–S15, 2000.
27. Hall DA: Should etching be performed as a part of a vital bleaching technique? *Quintessence Int* 22:679, 1991.
28. Papathanasiou A, Kastali S, Perry RD, et al: Clinical evaluation of a 35% hydrogen peroxide in-office whitening system, *Compend Cont Educ Dent* 23:335–346, 2002.
29. American Dental Association Council on Scientific Affairs: Report on laser bleaching, *J Am Dent Assoc* 129:1484, 1998.
30. Shethri SA, Matis BA, Cochran MA, et al: A clinical evaluation of two in-office bleaching products, *Oper Dent* 28:488–495, 2003.
31. McCloskey RJ: A technique for removal of fluorosis stains, *J Am Dent Assoc* 109:63, 1984.
32. Croll TP, Cavanaugh RR: Enamel color modification by controlled hydrochloric acid-pumice abrasion: Part 1. Technique and examples, *Quintessence Int* 17:81, 1986.
33. Croll TP: Enamel microabrasion for removal of superficial dysmineralization and decalcification defects, *J Am Dent Assoc* 120:411, 1990.
34. Dumfahrt H, Schaffer H: Porcelain laminate veneers: a retrospective evaluation after 1–10 years of service, *Int J Prosthodont* 13:9–18, 2000.
35. Friedman MJ: A 15-year review of porcelain veneer failure: a clinician's observation, *Compend Cont Educ Dent* 19:625–636, 1998.
36. Hashimoto M, Ohno H, Kaga M, et al: Resin–tooth interfaces after long-term function, *Am J Dent* 14:211–215, 2001.
37. Meiers JC, Young D: Two-year composite/dentin durability, *Am J Dent* 14:141–144, 2001.

38. Calamia JR: Etched porcelain facial veneers: a new treatment modality based on scientific and clinical evidence, *N Y J Dent* 53:255, 1983.
39. Friedman MJ: The enamel ceramic alternative: porcelain veneers vs. metal ceramic crowns, *CDA J* 20:27, 1992.
40. Stangel I, Nathanson D, Hsu CS: Shear strength of the composite bond to etched porcelain, *J Dent Res* 66:1460, 1987.
41. Bayne SC, Taylor DF, Zardiackas LD: *Biomaterials science*, Chapel Hill, NC, 1991, Brightstar.
42. Haywood VB, Heymann HO, Kusy RP, et al: Polishing porcelain veneers: an SEM and specular reflectance analysis, *Dent Mater* 4:116, 1988.
43. Cooley RL, Burger KM, Chain MC: Evaluation of a 4-META adhesive cement, *J Esthet Dent* 3:7, 1991.